

**RESEARCH METHODS IN DEVELOPMENT
COMMUNICATION**

EXPERT COMMITTEE

CHAIRPERSON: Prof. Nageshwar Rao, Vice Chancellor, IGNOU, New Delhi.

External Experts

Prof. J.S. Yadav
Former Director, IIMC, New Delhi

Prof. Biswajit Das
Founder Director, CCMG, Jamia Milia Islamia, New Delhi

Prof. Pradeep Krishnatrey
Former Director, Research and Strategic Planning, Johns
Hoppins University, Centre for Communication Programme
New Delhi

Dr. Archana Kumar
Associate Prof., Dept. of Development Communication &
Extension, Lady Irwin College, University of Delhi

Internal Experts

Prof. B.K. Pattanaik
SOEDS, IGNOU

Prof. R.P. Singh
SOCE, IGNOU

Prof. Jagdish Sharma
SOTST, IGNOU

Prof. Parmod Kumar
SOH, IGNOU

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Faculty of SOJNMS

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Dr. Usha Chandar
Research Officer, IGNOU, New Delhi

Dr. Ramesh Yadav
Convenor, IGNOU, New Delhi

PROGRAMME COORDINATOR: Dr. RAMESH YADAV

COURSE COORDINATOR: Dr. K. S. ARUL SELVAN

Course Editor:	Language Editor:
Dr. K. S. Arul Selvan, SOJNMS, IGNOU	Prof. Parmod Kumar, SOH, IGNOU
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Secretarial Assistance: Ms. Sunita Rawat | Sh. Amit Kumar Yadav

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Smt. Sumati Nair
Assistant Registrar
MPDD, IGNOU, New Delhi

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MDCE-006 INTRODUCTION TO COURSE: RESEARCH METHODS IN DEVELOPMENT COMMUNICATION

The field of mass communication has grown significantly during the past few decades. As a student of communication, and as a future practitioner in the field, it is essential that you are exposed to communication research methods especially in the areas of development sector.

The **Course MDCE-006** has been developed as foundational course in a coherent, logical, and systematic way to take you through the process of doing step by step research in this field. It is presented in Five Blocks comprising 19 units covering a range of issues in communication research methods and its application in development sectors. Depending on the thematic unity of the content, some blocks have five units, while others have four/three units – each building upon the previous one.

Block No. 1: Communication Research addresses the basic concepts in research. Unit 1 introduces you to the concepts – nature and scope of communication research. Unit 2 describes how communication research is classified in terms of its essential qualities and purposes; Unit 3 takes you through the process of formulating research problems, questions, and hypotheses. Unit 4 explains the process of sampling, so essential to design a study and Unit 5 deals with various aspects of the review of literature.

Unit 1: Research: Concept, Nature & Scope introduces Research, its nature and scope in a broad way. The central concepts, perspectives and various fields in which communication research is carried out, have been discussed in this unit.

Unit 2: Classification of Research describes varied research approaches based on design, stage, nature, location, approach and elements in communication. It helps you in deciding the type of research you would like to undertake, depending upon the research objectives and questions that you wish to explore.

Unit 3: Defining & Formulating Research Problems discussed the process of selecting a research problem, defining the terms of enquiry – research questions, objectives and hypotheses to help you in defining and formulating your own research problems.

Unit 4: Sampling Methods looks at sampling as part of the research process which involves the selection of elements to represent the population being studied.

Unit 5: Review of Literature discusses the need and importance of literature review in research work, the process of selection and organisation of the material and how to write a literature review.

It is advisable to study the units in a sequence as each builds upon the previous unit, and subsequent blocks are dependent on this introductory block. We hope that you will enjoy studying this Block and develop a good understanding of the concepts and perspectives discussed.

Block No. 2: Research Methods- Quantitative has four units exploring quantitative research in mass communication. Unit 6 teaches you how to find

primary and secondary data. Unit 7 describes the Survey Method and shows you the fundamental aspects of carrying out a survey. Unit 8 details how content analysis – a method pioneered by communications researchers is undertaken and Unit 9 explains how to design and carry out an experimental study.

Unit 6: Data Collection Sources describes the detailed processes of searching primary and secondary data and explores various data sources available in India. It explains how you can undertake the Internet and library search effectively and efficiently.

Unit 7: Survey Methods discusses the characteristics, types and applications of survey method. It explains the data collection tools and the process of conducting survey, including planning, sampling, designing and pretesting the research instrument, and analysis of the data collected.

Unit 8: Content Analysis focuses on Content Analysis, a method extensively used in mass communication to analyse how information is covered in media. The unit explains how content analysis is undertaken in a systematic manner – right from the statement of the research problem to arrive at conclusions and inferences.

Unit 9: Experimental Methods deals with experimental research which is very popular in psychology and social psychology but is relatively less used in media research. The unit explains the nature and scope of experimental method and how to design and undertake and experimental study.

We hope that this Block will give you the necessary orientation on data collection sources and quantitative research methods and enable you to use these methods in your own research projects in a systematic manner.

Block No. 3: Research Methods- Quantitative focuses on qualitative research methods. Unit 10 discusses Interview and Unit 11 details the case study approach and tools. Unit 12 teaches you how to undertake observation studies in the field; while Unit 13 explores the field of semiotics, where signs, symbols and meanings have to be understood as part of qualitative analyses.

Unit 10: Interview Techniques discusses the topic of Interviews as research tools. Interviews can be part of both quantitative and qualitative methods, and the qualitative aspect of interviews is highlighted here.

Unit 11: Case Study describes the strengths and limitations of this research methods and explains how to plan, design and conduct a case study.

Unit 12: Observation Methods describes how systematic observation is undertaken and data from observation are recorded and analysed.

While each tool is dealt with in a separate unit, it is important to note that the tools may be used as stand alone, i.e. only one tool for a study; or together in a complementary and supplementary way. Much of mass communication research uses several tools in a research study. The purpose is to better understand a phenomenon and to collect as comprehensive data as possible so as to make an educated and valid analysis of a phenomenon.

We hope you will enjoy working through the units and use the qualitative methods, tools and techniques of mass communication research discussed in this Block.

Block No. 4: Analysis & Interpretation is devoted to the important areas of data analysis and interpretation. Unit 14 introduces you to elementary quantitative statistics in communication research. This unit must be seen alongside Unit 15, which, while looking at qualitative research, also explains how to organise data for analysis and what statistical techniques you should use and when. The final Unit 16 takes you through the process of report writing, showing you how a report should be written in a systematic manner.

Unit 13: Basic Statistical Analysis discusses different tools of descriptive statistical analysis in detail with the help of exercises. You may find this unit slightly longer than the others. This is for two reasons: the first is that we are using a lot of numbers and a few formulae; the second being that, for mass communication students, explanations need to be simpler and clearer. It is especially important that you do the exercises in order to understand the concepts explained.

Unit 14: Data Analysis focuses on the concept of data analysis in research and the process of organising and coding of data for effective analysis. It also explains how to distinguish between quantitative and qualitative data analysis so that you can apply appropriate data analysis techniques in your research work.

Unit 15: Report Writing explains how a report should be written and presented; how the content should be organized, what are the sub-sections; what goes into each sub-section; and what are current practices in style and formats. Also included is important information about plagiarism. In the final analysis, this report is what is presented either for evaluation by an examiner, for a panel of experts on the subject, and even for other researchers to read.

After reading this Block, you may go back and refer to any unit when you are in need of help and also for a better understanding of the subject.

Block No. 5: Development Research consists of four units namely Basics of Development Research, Methods of Development Research and Development Research Applications

Unit 16: Basics of Development Research introduces meaning and concept of development research, purpose of development research, types of development research, approaches to research in development work, conditions for development research, key principles of development research, community consultation and development research. This unit also discusses conventional research vs development research, the roles required of a development researcher, outcome of development research and limitations of development research.

Unit 17: Methods of Development Research gives an account of various methods of development research namely, direct observation, interviewing, focus group discussion, case study method and processes of development research.

Unit 18: Development Research Applications discusses application of development research, appraisal of existing situation, need assessment, stakeholders' analysis, formulation of development project, monitoring of development projects and participatory evaluation.

Unit 19: Monitoring and Evaluation explains the importance of monitoring and evaluation methods in any development project. This unit gives you an orientation an tools and techniques of monitoring and evaluation process, in order to understand the ongoing development works.

We are sure that you will use all this orientation into research theory and practices to further the goals of development communication.



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Block**1****COMMUNICATION RESEARCH**

UNIT 1**Research: Concept, Nature & Scope**

UNIT 2**Classification of Research**

UNIT 3**Defining & Formulating Research Problems**

UNIT 4**Sampling Methods**

UNIT 5**Review of Literature**

UNIT 1 RESEARCH: CONCEPT, NATURE AND SCOPE

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Research: Concept and Role
 - 1.2.1 Definition
 - 1.2.2 The Central Concept
 - 1.2.3 Growth and Development
 - 1.2.4 Importance of Research
- 1.3 Research: Nature and Characteristics
 - 1.3.1 Nature of Research
 - 1.3.2 Characteristics of Scientific Research
- 1.4 Purpose of Research
- 1.5 Scope of Communication Research
- 1.6 Let Us Sum Up
- 1.7 References and Further Readings
- 1.8 Check Your Progress: Possible Answers

1.0 INTRODUCTION

Research in mass communication has developed into a full-fledged area of academic activity. Today it is multi-faceted and covers a wide variety of topics. Often, it is a product of inter-disciplinary research drawing upon research methods adopted by different social sciences like sociology, political science, psychology and others. From effects research to textual analysis to critical discourse analysis to semiotics, a wide spectrum of research methods are applied to various communication situations.

At this stage it is important for you to understand that research is a way of thinking as well as a skill which can be acquired. Research involves a holistic critical examination of an issue; identification and explanation of the principles that govern a given process; a search for solutions; and a statement of predictions. This is done for promoting knowledge and building theory and it is also done to find solutions to real life issues and problems in industry and society.

Sometimes we are asked to conduct research by an organisation. The first question that comes to mind is Why? And How? What is the purpose for which we are undertaking this extensive exercise? Is it to build on the theories or to address a specific problem that an organisation is facing? These questions are important and to address them, it is important to know that research is a logical process, which makes it necessary that we understand the broad concepts, and the nature and purpose of research. This block is an introduction to a complex field of communication research, and is designed to set you on an exciting journey in mass communication research.

1.1 LEARNING OUTCOMES

After working through this unit, you should be able to:

- discuss the concept, role and importance of research;
 - describe the nature and characteristics of research;
 - differentiate between Pure and Applied Research; and
 - outline the purpose and scope of mass communication research.
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1.2 RESEARCH: CONCEPT AND ROLE

In this section, we will focus on the core issues that characterise research. We will also trace its growth trajectory driven by various socio-political factors. Constant Research is even more important to enhance development communication.

1.2.1 Definition

Research can be defined as a careful investigation or inquiry especially through search for new facts in any branch of knowledge. Here, the term 'search' is important as researchers are looking for answers and can never be sure where they will find them or what they will find. In fact, the word 'research' can be seen as a composition of two syllables, *re* and *search*, suggestive of a continuous search for truth and disrespect for the status quo.

Within social sciences, we find the same emphasis in research on inquiry for theory building and on testing out of and verification of existing theories. In the past century, communication as a subject has drawn its theoretical basis from both physical and social sciences and has emerged as a discipline that is both multi and interdisciplinary. Communication research draws from the definitions of social scientists but focuses on any "research discipline that can shed new light on mass communication processes, effects, institutions and institutional change" (Westley and Stempel, 1981).

1.2.1 The Central Concept

Research is characterised by dynamism and the central concept is that of 'tentative truth'. A researcher aims to explore or uncover something that is, in some way, unnoticed, hidden, secret or problematic.

There is also a symbiotic relationship between theory and research; each is part of the process as also the result of the other. Findings from research feed into theory; which in turn forms the basis for further enquiry. Every theory in communication that has been propounded till date is a result of numerous revisions and refinements, many at the hands of the original author him/herself. Some theories have stood the test of time while others have not- thus have been rejected. For instance, as discussed in Course MJM-030, the Bullet theory of the early 1900s made way for the Two-step Flow hypothesis of the 1940s; Everett M. Rogers' Diffusion of Innovations theory of 1971 saw a reversal barely five years later making the original premise of the theory irrelevant. Other theories of media importance in development were also set aside when repeated results showed

that this was not the case. Thus, both theory and research are necessary to validate or refute each other.

1.2.3 Growth and Development

Historically, four major social forces were responsible for the development of research in mass communication - World War I revealed the power of media as a tool of propaganda. The Hypodermic needle (Bullet) theory of the early 1900s suggested that whatever message was pushed through the media was received as it is; by the audience who seem to exercise little or no choice about the content.

The second major force was the Manufacturers' /Advertisers' Lobby. Mass production of goods had taken place during the wars and once the wars were over, manufacturers were looking for buyers. In a market scenario, where supply exceeded the demand, they recognised the importance of message effectiveness and media planning in reaching out to the target buyers. They demanded hard facts regarding circulation figures, demographics of readers etc. from the publishers so that they could tailor and place the advertising messages accordingly. So research began to be conducted on the science and art of communication.

The third force was that of the Publishers' Lobby. From an earlier time, when the printer was also the publisher, the one-man printing establishments grew into an industry. Publishers recognised the profit in selling newsprint space to advertisers. They also realised that more readers meant more advertisers. Research began to be conducted into, 'who the readers were and what they wanted' - resulting in extensive readership and audience surveys.

The fourth social force was Public Concern. As media became more and more powerful and intrusive, people began to be more concerned about media impact on their personal and professional lives as well as on society. Thus began institutional and public policy research in communications.

Phases of Development: R.D. Wimmer and J.R. Dominick in Mass Media Research have identified four phases into which research in a particular medium is divided. These phases are not so much chronological but they are interlinked.

In Phase 1 there is an interest in the medium itself. What is it, how does it work, what technology does it involve, how is it similar to or different from what we already have, what functions or services does it provide, who will have access to the new medium, how much will it cost? All these questions can be asked of the Internet, Mobile Telephony and other forms of New Media.

Phase 2 begins once the medium is developed. Here the focus is on the uses and users of the medium. For instance, what are the extent and nature of activities possible with the use of New Media by different users? What is the demographic profile of the user? Are the young using it more? Do women use the Internet more for socio-communicative purposes as compared to men who use it more for informative purposes? Is the Internet a functional alternative to traditional media? Is it being used in ways that were not originally intended?

Phase 3 investigates the social, psychological and physical effects of the medium. How much time is spent in using the medium? When does the use become abuse/misuse? How much does it help? Do the shy take advantage of online anonymity

to express themselves more freely? Does it reduce the necessity of physical activity? Can it be combined with other media to make it more useful? Does it change people's perspective? Does it enable a level playing ground in business?

Does it bring in more players with different agendas? And so on.

Phase 4 focuses on how the medium can be improved - either in terms of usage or technology, for instance, by improving transmission and reception, by increasing the penetration, by making it more valuable through improvement in content etc. This phase has been largely driven by the private sector. Intra and inter-media competition for a larger profit share has ensured that the focus remains on how to retain audience share and increase it by attracting it away from the competition. Consumer comfort becomes the business mantra for success.

1.2.4 Importance of Research

As discussed above, for us the importance of research lies in its implications - whether it is meant for knowledge creation or solving real world problems.

- Research is a critical input into the development, design, production, delivery and evaluation of courses and content; while on the other hand, findings from research must feed into the theory of this multidisciplinary and even epistemological enquiry. The perspective must then become total and each element must be examined in association with each other, rather than in isolation at each stage of the process.
- Research is an important prerequisite for understanding social order. Social phenomena are complex. There is constant interaction of diverse influences environmental, psychological and social; on human beings, as individuals, members of a social group, and as a society at large. Therefore, research is needed to investigate both isolated patterns of behaviour as also to identify set patterns of behaviour exhibited by social groups and societies.
- Through a systematic interrelation between disparate facts, research allows us to understand, generalise, predict, and manage social behaviour.
- Research generates voluminous social data which help the formulation of social policies, planning and priorities. In a culturally diverse country like India, such data are critically important for judicious use of resources and coherent development of a rapidly changing society.
- Research is both cumulative and a self-correcting process. Its importance lies in the fact that, if done well, it can be trusted. The conclusions arrived at in a research study are firmly tied to reality as they are based on a logic born out of observation and measurement.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Define Research.

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2) What is the central concept of Research?

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3) List four reasons for the growth of research.

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4) What is the importance of research?

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1.3 RESEARCH: NATURE AND CHARACTERISTICS

After an overview of the concept, role and importance of research, let us understand the nature and characteristics of research.

1.3.1 Nature of Research

As mentioned earlier, the basic nature of research is to advance knowledge and seek solutions to problems. To do this, we start with simple questions. For instance, in journalistic practice, the basic questions are: who, what, why, where, when and how. In research, these questions are addressed in a more systematic, reliable, testable, and replicable manner.

In practice, all the questions are mixed together and it is difficult to isolate one from the other when dealing with human behaviour and social phenomenon. In research, these are isolated and studied in depth - separately as well as together.

The basic premise is that any issue/event/phenomenon can be studied, subjected to appropriate systematic, objective scientific procedures and conclusions can be arrived at that can preferably be generalised to the population. Such results and conclusions should also be amenable to replication as the search for knowledge is conducted with a defined set of rules and procedures that are commonly understood and shared by all sciences.

1.3.2 Characteristics of Scientific Research

Research needs to be done in a systematic and logical way, following a series of steps and sets of protocols. Without such procedures, research cannot be considered scientific. Let us look at the basic tenets of research:

Research is empirical

The word 'empirical' comes from the Greek word 'empiricism' which means experience. Researchers are concerned with what is observable and potentially measurable. For instance, the problem of declining circulation figures can be investigated, identified and measured in numerical terms. With the findings, solutions to address declining circulation can be suggested.

Since research depends on observations, the factors or variables under study are defined in operational terms (as operational definitions) so as to be either observed or measured i.e. observable behaviour patterns that can be connected to represent abstract (covert) concepts. For instance, if we argue that modernity in an individual is associated to educational levels, the variable of education can be easily measured in terms of the educational degree possessed such as matriculate, graduate etc. The variable of modernity, being more abstract and complex, can be measured in terms of how the respondent tests on a readily available scale, such as K.S. Yang's Multiple Modernity Scale or on a scale created by the researcher to measure the responses on concepts related to modernity.

The basic aim of research is to arrive at an understanding of why a phenomenon occurs or is likely to occur. Therefore, we use concepts to organise and make sense of things. In many ways they are like labels that help us in structuring our observations. They also enable us to measure the associations and correlations between and among various concepts. For instance, to study the relationship between violent behaviour and viewing of violent TV content we may ask- Does televised violence 'cause' violent behavior? This is a very complicated issue and researchers have been arguing for decades on how to define violence and how to determine whether the effect of watching televised violence is 'significant'. Depending on how the researcher operationally defines terms, the respondent behaviour can be observed, measured, and results and conclusions arrived at.

Research is objective

All human beings have their own attitudes and perspectives on different aspects of life. The attitudes and perspectives are a result of the educational and socialisation process in the culture in which we have grown up. Such attitudes can affect the way in which we conduct research. Therefore, in order to be accurate in research, it is important that the research process be objective and as free as possible of biases and pre-conceived notions, following the scientific method.

The scientific method presupposes ethical neutrality on the part of the researcher and tries to rule out personal biases and judgments by laying down explicit rules and procedures in place for the study. Each approach or method of research may have different ways of reducing bias and prejudice, but at the basic level, rules and procedures exist. Among the most important is that the data collected during the course of a research study should speak for itself - rather than be tilted in favour of a researcher's opinions. Results are often not as we expected them to be, but the facts must stand on their own and reflect an objective finding which is as free of bias as possible. As the noted psychologist B.F. Skinner wrote in *Science and Human Behaviour*, "Research projects do not always come out as one expects, but the facts must stand and the expectations fall. The subject matter, not the scientist, knows best."

The 'creative' part of research is usually limited to the identification and formulation of the research problem. But once the methodology is framed, its implementation should be as per the laid down procedures as the methodology of a study determines the reliability and validity of the findings and in turn contributes in knowledge generation. For instance, a researcher, based on his observation, that majority of people discuss the stories covered by a specific newspaper may hypothesise that it is the largest selling newspaper, but it is only the collection of primary or secondary data that will support or reject the hypothesis.

Objectivity is also ensured by framing operational definitions as mentioned earlier. This, not only establishes that the researcher has used logic and reason to study a particular phenomenon but also facilitates replication of the study by other researchers on the same or different parameters. Operational definitions also help in defining the meaning of a term for the specific purpose of the given research effort. For instance, a researcher may define violence as a physical act resulting in bloodshed and then analyse a particular programme to determine how violent it is. Another researcher may define violence as verbal abuse and analyse the programme accordingly. Still other researchers have the choice to replicate the studies of these two researchers by using the same parameters or conduct independent studies using their own definitions. The end result will contribute to enhancement of knowledge and a holistic study of violence along various dimensions. You will read more about defining operational definitions in subsequent units.

Another method to ensure objectivity is through 'triangulation' - when more than one researcher observes the same phenomena for the same study or when more than one set of data are compared to determine the final result. This ensures that individual biases are ruled out and a consensus is arrived at. Triangulation can also be incorporated into research by using more than one data collection tool to observe/measure the same phenomenon for cross verification.

Research is systematic and cumulative

Not only is research systematic, it also builds on existing knowledge. No research study stands alone, nor does it rise or fall by itself. Previous studies are always used as building blocks or launch pads for current research. The idea is to strengthen existing theory or to revise it by focusing on the gaps or shortcomings of previous research. The examination of previous research, also called "Review of Literature" comes before the formulation of the terms of enquiry and the framing of appropriate methodology for the collection and analysis of data.

Research may start with a single, carefully observed event and progress ultimately to the formulation of theories and laws. These theories, in turn are applied to other single carefully observed events and compared with previous results. Following a constant cycle of testing and verification, theories are built, tested, rejected, and rebuilt.

Research is predictive

Science is concerned with relating the past/present to the future. Research strives to develop theories because they are useful in not just understanding the past/present, but in predicting future behaviour in different situations.

Research is public

Researchers, especially those in the academic sector, work publicly. Research in the public domain helps to build theory and is necessary for academic peers to compare, test, retest, and validate findings. Even private sector research, if it has implications for the public, as for instance in pharmaceuticals, has to be publicly available.

The public nature of academic research implies responsibility in publication. The published report should include all pertinent details so that the readers are in a position to judge whether the findings are reliable and valid enough to be used as a building block for further research or as the justification for a policy decision. Even where research findings are not made public but used in-house, the researchers involved have to reveal the methodology used in data collection and analysis, as too much hinges on the reliability and validity of the findings.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What are the characteristics of scientific research method?

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2) What is empiricism?

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3) How is objectivity achieved in research?

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1.4 PURPOSE OF RESEARCH

Earlier in the unit, we touched upon the symbiotic relationship between theory and application. On the one hand, research is a critical input into the development, design, production, delivery and evaluation of content; while on the other hand, findings from research feed into the theory of this multidisciplinary field of communication. The perspective then becomes total and each element is examined in association with each other, rather than in isolation at each stage of the processes.

If research has no implication for theory, if it lacks even a minimum of *scientific method* and explanatory power, it should not be undertaken. At the same time, if research in the context of national social and economic development cannot provide a clear, direct, and unambiguous line between results and decision alternatives, it lacks *pragmatic validity* and similarly should not be undertaken.

Research can be classified into pure and applied research:

Pure Research

Pure research is when knowledge is pursued purely for intellectual curiosity. It is undertaken for the satisfaction of knowing or understanding, regardless of whether they have any practical application for the present or the future. The emphasis here is on discovering, developing and testing theories. For instance, a set of propositions that showed how and why audiences considered some news stories more significant than others was eventually termed as the Agenda Setting Theory. Pure research is also concerned with the development and refinement of research procedures, techniques and other aspects of research methodology. For instance, the development of a standardised scale to measure Internet addiction or the development of a more sensitive instrument to measure Emotional Quotient (EQ) would help the research community conduct comparable research across different groups in different cultures.

Applied Research

Applied research is conducted to solve practical problems of efficiency and profit. In print media, applied research comprises of readership studies - the demographic and psychographic profiling of readers and non-readers. Need gratification studies are conducted to determine who is reading what and why. Other kinds of studies are done to determine how variation in presentation (typography, makeup etc.) influences understanding of content; management studies. These are some examples of applied research in mass communication.

In electronic media, ratings and non-ratings research is conducted to determine the popularity of a programme, demographics and psychographics of its audience, formulation and implementation of creative strategy, programme scheduling, host/anchor popularity, etc.

Applied research is also conducted in the areas of advertising and public relations. In the former case, research is conducted along cognitive, affective and behavioural dimensions i.e. how advertising copy should be crafted so that it provides both information and entertainment while at the same time prompting action in the form of product purchase/attitude change. Research is also conducted to determine the right media mix - markets, budget allocations, choice of media channels, message distribution and advertising mechanics such as purchase of time units, print space etc.

In Public Relations, environmental monitoring programmes are conducted to observe trends in public opinion and socio-political events that could have significant impact on an organisation. Public relations audits are conducted to determine and assess image management activities; communication audits are done to assess the impact of internal and external communication tools such as press releases, in-house publications. Further, social audits are conducted to measure an organisation's social performance i.e. how well is it living up to its public responsibilities in a society.

It is important to mention here that it is not always possible to classify a given study neatly as one or the other. Pure research itself is an application of knowledge garnered from previous research and/or the researcher's own thought processes. Similarly, applied research is basic in the sense that it is true to a particular problem being currently studied. It is also difficult to make a distinction based on that one searches for knowledge and the other applies it or that one comes before the other. It is true that basic research often yields knowledge that has immediate practical utility but applied research may also yield the discovery of basic principles. For example, when the application of the Bullet Theory could not predict the outcome of the 1948 American Presidential elections, it was studied again and eventually paved the way for Two-Step Flow theory.

1.5 SCOPE OF COMMUNICATION RESEARCH

As we have already seen, communication research extensively borrows from anthropology, sociology, political science, economics, psychology and other disciplines. Communication research is about researching communication - human and technology mediated communication. Media and communication studies are one of the parts of human communication which includes media message creation, design of those messages, distribution of those messages, and consumption of media messages.

Media messages inevitably include use of technology. For example, you read newspaper, watch television, listen to radio, use mobile phone or see film in a cinema hall. All these activities consist of areas of research within communication research. Anything and everything about communication can be researched; who is creating communication, what is being communicated, how the communication is being sent and how is it being received, which communication is reaching the receiver and what are the effects of communication - all these are within the purview of communication research.

In the previous section, we have seen how communication research encompasses market research, film studies, media research, audience research, content research, and advertising research and so on. The basic elements of communication sender-message-channel-receiver-feedback can be researched individually or in combination with each other. Communication research includes researching each of them individually using suitable methods or examining the effects on each other. However, what specific aspects of each of these areas do communication researchers study?

Communication researchers study three things: Firstly, communication production, transmission and meaning making. Secondly they study content or form of communicative messages. Thirdly, communication researchers study functions and effects of communication messages. There are multiple branches of communication research.

When we link communication research to other social sciences, new areas or branches of communication studies emerge. Some of these are:

- Communication policy and law
- Organisational communication
- Cross-cultural systems

- Intra-personal communication
- Family communication
- Development communication
- Marketing communication
- Persuasive communication
- Political communication
- Health communication
- Instructional communication
- Public Relations

Within each of these sectors, like any other methods of research, communication research can be further viewed from quantitative and qualitative methods perspective. Quantitative methods encompass surveys, content analysis, experiments whereas qualitative analysis includes discourse analysis, ethnographies, case studies, semiotics, and rhetoric analysis and so on. You will read about them in detail in Block 2 and 3 of this course.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Name the disciplines from which communication draws its methods.

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2) What are the three things communication research studies?

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3) Mention some research methods used in communication.

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1.6 LET US SUM UP

The central concepts, broad perspectives and purposes, and the various fields in which communication research is carried out were introduced in this unit. The importance of communication research, either for expanding our knowledge or for solving specific problems was also discussed.

The relationship of communication research to other disciplines was explored at some length. We also discussed specific areas in the media industry where communication research has helped organisations to expand readerships, improve their product image and credibility. What communication researchers study in terms of the medium, the content and the meaning was explained.

This is the introductory unit of the course. Subsequent units will take you through a logical sequence of how to carry out communication research in much greater detail. Therefore, understanding the basic concepts presented in this unit is essential and you may read through the unit again, if you need to, so that you understand the full background of the subject.

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1.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Research is a “systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relations among natural phenomena.”
- 2) The central concept of research is ‘tentative truth’. The word ‘research’ can be seen as a composition of two syllables, *re* and *search*, suggestive of a continuous search for truth and disrespect for the status quo.
- 3) The four major reasons for the growth of research are the World Wars, Manufacturers’/Advertisers’ Lobby, Publishers’ Lobby and Public Concern.
- 4) The four phases of development in research are the medium, uses and users of the medium, effects of the medium and improvements of the medium.
- 5) Research is important for the evolution of civilisation, for promoting a dynamic social order that is free of myth and superstition. It enables us to exercise control of our environment by increasing our ability to predict our future. It also helps us to put our past into perspective as well as understand our present.

Check Your Progress 2

- 1) The scientific research method is empirical, objective, systematic and cumulative, predictive and public.
- 2) Empiricism is derived from ‘experience’ - something that can be observed and measured.
- 3) Objectivity in research is ensured by focusing on method which should be detailed and logically defensible.

Check Your Progress 3

- 1) Anthropology, Psychology, Political Science, and Sociology.
- 2) Communication researchers study the following:
 - Communication production, transmission and meaning making
 - Content or form of communicative messages
 - Functions and effects of communication messages.
- 3) Some methods used in communication research are surveys, content analysis, experiments whereas qualitative analysis includes ethnographies, case studies, semiotics, and rhetoric analysis.



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UNIT 2 CLASSIFICATION OF RESEARCH

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Classification of Communication Research
 - 2.2.1 Based on Design
 - 2.2.2 Based on Stage
 - 2.2.3 Based on Nature
 - 2.2.4 Based on Location
 - 2.2.5 Based on Approach
- 2.3 Study of Elements in Communication
 - 2.3.1 Communicators
 - 2.3.2 Media Content
 - 2.3.3 Distribution
 - 2.3.4 Audiences
- 2.4 Let Us Sum Up
- 2.5 References and Further Readings
- 2.6 Check Your Progress: Possible Answers

2.0 INTRODUCTION

Research is the Central point in the domain of Development Communication. In the previous unit, we discussed the concept, nature and scope of research. In this unit, we shall discuss various classifications of communication research. The communication process in terms of its elements such as sender-message/ channel-receivers-feedback can be researched at various levels. We shall explain how communication research can be examined using various lenses in terms of design, stage, nature, location, and approaches to research human communication.

How a given research project is done can be described in many ways. For instance, the way in which you plan to design your research is one way. Similarly, when you conduct your research or the 'stage' of research, i.e. before, during, or after a project, can be another way to classify it. A third way to classify research is by the reason for which it is done - for what purpose or objective is the project undertaken. A fourth way to classify research is by the location in which it is carried out - in the field or in a laboratory. A fifth way of classifying research is by locating it within a discipline, i.e. in agriculture, psychology, political science, or sociology, among others. Yet another way to classify research is based on the theoretical approaches with the field, i.e. structural, behavioural, or critical. How you approach a research project often defines the broad methodologies that are followed. Finally, we can also study four broad aspects of communication research- communicators, media content, distribution and consumption by audiences.

2.1 LEARNING OUTCOMES

After reading this unit you should be able to:

- analyse various classifications of research;
- describe possible approaches to communication research; and
- examine ways of doing research within media and communication discipline.

2.2 CLASSIFICATION OF COMMUNICATION RESEARCH

Communication research can be examined from various strands and there is no single way to classify it. Different experts in communication classify approaches to research differently. The ways of looking at it can be based on how you wish to examine it. The design, stage, nature, location and approach are some of the ways to classify communication research. Let us discuss each of them in detail. there may be some overlap. However, because despite the best efforts, communication research cannot be placed only in one classification. The ways of looking at communication research projects depend largely upon the objectives and expected outcomes of your research study.

For the purpose of this course, we shall follow the classification used by two well known communication scholars, Roger Wimmer and Joseph Dominick in their book “Mass Media Research: An Introduction” since their approach is most relevant for the studies you may be conducting. They have examined applications of communication research into print media, electronic media, advertising and public relations, Internet and effects research. Media research is a very significant part of communication research.

2.2.1 Based on Design

Research design is a philosophical framework whereas method is a more tangible way to undertake research process. If we examine communication research designs there can be broadly four categories - descriptive, diagnostic, exploratory, and experimental. Historical or archival methods too can be part of any of these four categories.

Descriptive

When we want to describe a phenomenon under study, it is called a descriptive design. Let us say, we want to describe how people watch television or what happens when people watch film in the theatre or how do children deal with the mobile phones of their parents - all these are possible descriptions of individuals interaction with media.

Diagnostic

This design refers to understanding cause-effect association or solving a problem on hand. For example, if there is drop in the viewership of a show or there is a sudden rise in the usage of mobile data, the researcher can find out the reasons by using appropriate research methods. Most of the diagnostic studies are applied in nature.

Exploratory

There are occasions when variables are unknown or the phenomenon under study is a new development. In such a case, the researcher can use an exploratory design. This design provides flexibility for study and helps in the process of

identifying variables for future research. Exploratory studies provide the basic information upon which we can sharpen our research questions, identify our variables and state our hypotheses. When preliminary information is not readily available, exploratory research is found useful.

Experimental

In Experimental designs, the researcher can control and manipulate the situation in order to observe a phenomenon under study. This type of research can be conducted in a laboratory, in an online environment or even in real life field conditions. Experimental research can be of many types which are described in detail in Unit 9 of this course. An experimental design is useful if you want to analyse the extent to which a film on HIV/AIDS helps improve knowledge about the disease among subjects exposed to the film. The difference between what they knew before they saw the film and what they learned after the film, can tell you how effective the film you showed them was.

Historical

Historical designs are based on archival material. Historical research is also being used in communication research. For example, if you study the letters between Pandit Jawaharlal Nehru and his daughter, Indira Gandhi, it will help you understand various aspects of political and social life of that era. Media histories, audience experiences, memory research are some of the examples of historical designs within communication research.

2.2.2 Based on Stage

You can take the context of time of communication as an aspect of research. Based on the stage of communication, research can be classified as pre-communication (Formative), during communication (Process) and post communication (Summative) research. The Development Educational Communication Unit of Indian Space Research Organisation (DECU-ISRO) Ahmedabad pioneered this approach for most of its development communication projects starting with Satellite Instructional Television Experiment (SITE) in 1975.

Formative Research

Formative research is undertaken before any communication project is created in order to define the audience - for whom the communication is intended; the content - what will be the content, and the presentation - how will that content be communicated what will be the language, and so on. You will need to make decisions about the locale or language or set design or clothing or social-psychological-economic milieu of the content. For example, if you want to make a film about Mahatma Gandhi, you will have to research about clothing of that period, sets, his writings, possible locales, audience understanding, what you wish to convey through your film and many such things. All these constitute formative research which then serves as information inputs into the programme planning.

The E-PG Pathshala project of Ministry of Human Resource Development, Government of India has courseware for MA in Communication Studies under Social Sciences. There are chapters on Formative-Process-Summative research as a part of their Development Communication paper. You can read about

formative research in detail on URL: <https://epgp.inflibnet.ac.in/ahl.php?csmo=24> for a better understanding of how formative research can contribute to making communication effective.

Process Research

Information is needed when a communication project is in progress. For example, there is a need to know what is being communicated in terms of content, how is it being received, what are the contexts of reception, and what do audiences do when they receive communication. Such questions are addressed by process research. You may choose to take a telephone survey and ask randomly which programme is being watched by people when a particular television programme is being telecast and the finding will reveal if the programme you were researching is being watched or not. Various scholars have studied the telecasts of *Ramayan* and *Mahabharata* serials on Doordarshan during 1980s and found that people remained indoors and roads were empty at the time of telecast. People offered prayers when they sat in front of television sets and treated the actors of these serials as demi-gods and even asked for their blessings. The findings of process research may help in correcting problems, making changes in content and format, and in general, fine-tuning the communication project as per audience needs.

Summative Research

Unlike formative and process research, summative research is undertaken after the communication programme is completed. There are many reasons for conducting summative research. Often, we need to know the extent to which a communication project met the objectives for which it was designed; how effective the project has been; was the money well spent; and finally, should the project be modified or stopped or continued in a different form. There may be mixed results and summative research helps to understand what succeeded, what failed, and why. For example, summative research conducted after SITE (mentioned earlier in this unit) helped in many ways for future planning. Current work in development communication in India has its roots in the research carried out during SITE. Usually for a communicator, summative research serves as an investment for next communication project.

You can read more about summative research on URL: <https://epgp.inflibnet.ac.in/ahl.php?csmo=24>.

2.2.3 Based on Nature

Research can be carried out for the purpose of policy or prior to production (Pre-production), during production, during telecast, post telecast. This classification of communication research may not be used for a specific programme but for a group of programmes. It may be undertaken by a communication agency or by third party agencies to define the larger goals of communication.

Policy

Research is undertaken to provide inputs for policy level decisions. For example, it can examine if people have stopped watching television? Is mobile phone the only way to reach out to young audiences? Are films in cinema hall being watched mainly by middle-aged people so the themes need to be serious in nature and not mere love stories? Communication planners can address such questions in order

to make decisions about their communication plans as part of large national programmes such as Rural Health Mission and so on.

Pre-Production

Similar to formative research for a particular communication project, pre-production research is undertaken prior to the production of a programme or series of programmes. Pre-production research, as the name suggests aims to decide matters related to communication, audiences and even message. Pre-production research includes deciding target audience, story, script, and production related decisions about locations-sets-costumes-casting-property, pilot testing, and so on.

Production

This type of research is done when production is in progress. You may be aware that while production is in progress, the communicator may be required to make changes in the original plans. It may be to do with resources as costs planned earlier increase, actors are not available, new things are added on the spot or there may be changes while executing the production or even practical considerations while work is in progress. Production research helps in keeping a planned process under check and can also help in monitoring of work.

Monitoring of Telecast

Similar to process research which we discussed earlier, telecast research is when the communication telecast is in progress. For example, producers of television entertainment programmes often rely on audience viewership figures for deciding whether to change the script, change characters, extend a entertainment serial, or even when to end a programme. For example, when a character in a popular serial died or was killed by the producers because the actor left the show, people came to the streets and demanded that he should be brought back. Researchers have also noted that people watching religious serials touch the television screen and pray to the god/goddess characters in the serial! Such findings were possible due to telecast research. Visit of film makers to theatres to ascertain the audience response and other tools such as Peoples' meters or Television Audience Measurement are used by broadcasters to monitor telecasts.

Post-telecast

Like summative research, post-telecast research is undertaken to examine how communication is received, what was audience feedback, should there be any changes in communication for future, and so on. Post-telecast research is usually used in electronic media in order to understand issues for the next production.

It is important to understand that research serves as an important investment for a media organisation. For example, if a newspaper finds its circulation has gone up after introducing a new feature page, they may start charging more for advertising; and their incomes would go up as a consequence. Further, post-production research for a particular project can become pre-production research for the next production. Similarly pre-production research which helps in production can be useful in examining production process and can contribute to post-production processes.

2.2.4 Based on Location

Research can also be classified in terms of location - field research, laboratory experiments or field experiments.

Field research

Field research as the name suggests is carried out in locations. Going to a village would become a necessity for an organisation which wants to know if Wi-Fi is available in a village, what is the strength of the signal, and what kind of facilities are available for men and women to access the Internet. Such kind of information cannot be collected from a laboratory or a bookstore. Field research is carried out where the activity is actually taking place - the field can be home or a work place or an educational institution of the respondent. Most of the social science research is generally carried out in the field because a laboratory is controlled and cannot describe the real life situation. Variety of methods are used in the field, e.g. surveys, observations, and other ethnographic methods described in detail in Block 2 and 3 of this course.

Laboratory Experiments

Experiments are conducted in controlled environments created by the researcher as per the requirements of the respective research project. There can be varied types of experimental designs employed depending upon the requirements of respective research. The most commonly used method is pre-post design or control group-experiment designs. Laboratory experiments are extensively used in medical and pharmaceutical research; but they are also used in communication research. For instance, an experiment can be carried out for the pilot testing of a TV programme or an advertisement to see how effective it is likely to be, when launched. If it is not effective, then it is cast aside. Lab experiments are also conducted extensively in educational communications, when the project managers want to see if the content of the programme is useful for the target audience or not.

Field experiments

Field experiments are those where the researcher goes to the field and conducts experiments in the field itself to observe an intervention/experiment being undertaken. You may study about SITE (Satellite Instructional Television Experiment), JDCP (Jhabua Development Communication Project) which are examples of field experiments to analyse the role of satellite technology for social change. These experiments have used research to design communication, made changes based on research and undertaken evaluations to ascertain influence of such experiments.

2.2.5 Based on Approach

As discussed earlier, communication research is a multi-disciplinary subject. It borrows research methods from anthropology, sociology, psychology, economics, and political science and so on. Based on the approach undertaken by the researcher, there can be four strands - Structural, Functional, Behavioural, and Cultural/Critical.

Structural

Structural or Structuralism borrows from sociology, anthropology, literary criticism, architecture, psychology and linguistics. Structure means interrelated parts of a system, e.g. how the communication content is structured. Philosopher Simon Blackburn (2008) defines structuralism as “the belief that phenomena of human life are not intelligible except through their interrelations. These relations constitute a structure, and behind local variations in the surface phenomena there are constant laws of abstract culture”.

In order to understand structuralism, let us take an example from language. In linguistics (language as science of signs) we examine how pattern of words communicate meaning. As you know “I love you” and “I love you?” do not mean same thing in spite of the fact that both the sentences use the same words. The meaning will change just by changing the sequence of words or putting a question mark or an exclamation mark. Similarly in visual media, the composition or the sequencing of visuals communicates the message. The structural approach to research in communication examines how shots have been ordered in the film, how sequence of words in radio is changing the presentation, what has been the presentation in television show and how different genres of programming get influenced by structure. The Structural approach examines sequence, grouping, and patterns etc. and the change in sequence can lead to change in meaning in the context of media communication.

Language is a sign system of communication in human society and tells us many things about its speaker hence words spoken by a character can reveal a lot about him/her. Researchers using structuralism examine structural elements of the message in order to reach their conclusions. For example, researcher analysing the portrayal of women in films will examine camera angles, length of the shots, frames, the sequence of presentation, the relational difference in men versus women and so on. If certain parts of women’s body are focused upon more; then a different meaning will be attributed to such a portrayal.

Functional

This research classification examines what are the functions of communication. Uses and gratification theory is an example of functional approach to communication research as it examines the ‘why’ of communication. Why do people watch television, why do they keep watching the same film number of times, why do people keep listening to music even while they are on a busy road. Uses and gratification provides a useful framework to understand the ‘why’ of media consumption. The theory notes that there are: cognition, diversion, social utility, affiliation, expression, and withdrawal functions of media use by individuals. To take the above example of portrayal of women in films, if the researcher is using a functional approach, s/he will examine- what is the character given to women in the story line, how does her character take the story further, what the woman does in the film- is she within or outside the house- what is she doing- what kind of clothes is she wearing- what is she speaking and so on. If the functions given to women are ornamental or passive, then it may suggest that the portrayal of women in that particular film is cosmetic in nature and not significant.

Communication is also considered a behavioural science. Communication helps the receivers know about the issues, help in changing their attitudes about the issues and finally change their practices. But it may not always be in that order. For example, people are aware that 'smoking is injurious to health', but even then they smoke. The social awareness campaigns generally use fear appeal to stress that you will get cancer because you smoke. Now, if all the viewers who saw that campaign get impacted by the communication, everyone should have quit smoking but that does not happen and only a few people get influenced by the campaign. A behavioural approach on the importance of 'No Smoking' campaign will examine which members quit smoking after being exposed to the campaign and why.

Cultural/Critical

Cultural approach, as the name suggests, indicates signs, values, attitudes, beliefs, assumptions, language, even clothing, accessories, food etc. in a given society. Cultural approach in communication research looks at communication from the point of view of socio-cultural processes. Borrowed from anthropology, this approach to research examines communication as socio-cultural process and explores ritualistic processes due to communication. This approach acknowledges heterogeneity, diversity and the dynamic nature of communication process.

Cultural studies, unlike cultural theory, come from critical approach. Cultural studies relate to the study of ideology, culture, social structure and power and borrow from Marxism, feminism, race, culture, film theory, post colonialism, political economy, history, philosophy and so on.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What areas are included in communication research?

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2) List the types of communication research based on the nature and stage of programme.

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3) What is the difference between formative and pre-production research?

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- 4) What is the difference in structural-functional-behavioural and cultural approach to communication research?

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2.3 STUDY OF ELEMENTS IN COMMUNICATION

Communication research also includes researching various elements of communication including communicators, media text, distribution process and audiences. The communicators include the owners, creators, content producers, actors or technicians and so on while media text includes - film, television programmes, newspaper articles, radio programme or any media content. Distribution covers how and what reaches consumers/audiences/receivers of communication. The most important and challenging dimension of media communication is the audience. Consumption of communication in terms of its receivers /audiences- how communication is being received, effects of communication, consequences and context of consumption also need to be researched. Let us discuss each of these elements in detail.

2.3.1 Communications

Communication in a mass media environment is designed and executed by communicators. You watch television shows, play games on mobile, listen to radio programmes, watch films either on television or in theatres; all of these are created by communicators who work in teams, led by a team director. But as a researcher, it is interesting to find out how these productions get made. Communicator studies can be labelled as those in which institutions as media communicators cannot be individuals but are organisations/agencies/institutions or even media conglomerates.

Within communication research traditions, communicators can be researched using varied methods. For example, you can do in-depth interviews of the psychological or sociological characteristics of filmmakers to understand their perspectives. One can also study the management practices in television channels and the ownership patterns of media by either politicians or by large industrial houses to understand how their views influence the content of the media. You can also examine the language spoken by film directors on the film sets or study how actresses are treated on television sets and so on.

Communicator researches help us understand who creates the communication and under what conditions. There can be studies of policy, media economics, or political economy approaches to examine media institutions. There can also be managerial, organisational, financial studies in order to understand the working of media institutions.

2.3.2 Media Content

Media content is another important aspect of research - it can be television or radio programmes, newspaper articles, internet content, or films which make up

media text. A study of media content provides us information about the society from where the text has originated or society for which the text is intended, and being consumed. For example, Indian films are popular across the world. However, their popularity is not defined by who created the films but the way the codes, values and narratives are presented transcending the geo-political boundaries.

As a researcher you can use methods like content analysis or semiotic analysis to examine media content. Text analysis, literary analysis, semiotic analysis, message system analysis are some of the examples of communication material research. Study of text also helps you understand what is being communicated and who is being portrayed and how. For example, you can study how much space given in a newspaper to issues relating to senior citizens. By analysing the space, you can arrive at the importance of senior citizens' issues. You can also study how much time is given to politics in a news bulletin and compare that with the time given to health related issues, to determine the priority a television channel places on those subjects. What will you conclude, if over fifteen days, the major news bulletin of a news channel gives 60 per cent of its time to political news, 15 per cent to sports; 15 per cent to entertainment; and only 10 per cent to other newsworthy items?

2.3.3 Distribution

As you are aware, mass media reach audiences through various distribution channels. Understanding how communication is reaching the audience or how communication is being sent will help you to understand the ecology of communication. For example, newspaper distribution in India is still a complex process. However, the newspaper readership and consumption is linked to distribution patterns. Analysis of the distribution patterns and the impact of online availability of newspapers will help you understand the status of newspaper business in a given society.

2.3.4 Audiences

Receivers or audiences are the most important in the process of communication. As opposed to earlier conceptions of audience as individuals, literature now recognises them as plural and 'audiences'. Mass media audiences can be radio listeners, television viewers, newspaper readers, filmgoers, mobile phone users and so on. Since mass media audiences are heterogeneous and scattered it is important to research them. Researching audiences helps you understand if the communication reached them, how they received it, what the effects of communication were, and if any changes happened because of the communication in the lives of the audience.

You have studied basic communication process of sender-message-receiver-feedback in earlier units. Communication research has to examine how communication messages are received. If there is no receiver for the message, the process of communication research is not complete. The other examples of research related to receivers are audience research and reception analysis. The earlier traditions of research have effects tradition which viewed media as all powerful impacting each member of the media audiences. Today, post-effects traditions, uses and gratification, cultivation research, reception analysis view audiences as more active and creators of meanings.

Activity-1

Design a research study using two key words; “Women” and “Media”. Select the type of study you wish to design and the approach adopted, drawing upon the content of this unit.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Which elements of communication can be researched?

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2) Is it possible to design a research study that can examine all the elements of communication process? If so which approach will be useful for such study?

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3) Briefly mention the shift in the way audiences are perceived in communication research over the years.

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2.4 LET US SUM UP

In this unit you were exposed to various classifications of communication research. It was discussed that communication research can be examined using the lenses of design, stage, nature, location, and approaches to research communication. Communication research is also undertaken on the elements of communication; sender-message-receivers as well as feedback or effects of communication.

By now you would have understood that research problems can be studied using varied approaches and can be designed depending upon your research goals and questions that you wish to explore through your study.

In the next unit, we shall discuss defining and formulating a research problem. Keeping the classification of research in the background will help you to define your own research problem.

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2.6 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Communication research covers a wide spectrum of areas including media message creation, design of those messages, distribution of those messages, and consumption of media messages.
- 2) Based on the: Nature of programme: policy, pre-production, production, telecast, post telecast.
- 3) Stage of research: formative; process and summative research.
- 4) Formative research is about goals of programme, audiences, programme content and decisions about transmission /distribution whereas pre-production research is a micro exercise and is mainly decision making with reference to the particular production in terms of story, locations, script breakdowns, sets, clothing, properties and so on.
- 5) Structural approach to communication research focuses on structure of content whereas functional approach examines what is the function of that communication for audiences. Behavioural approach examines change in practices amongst members of audiences and cultural (which is also known as critical approach to communication research) views process of communication as an activity/process that impacts power and questions relationship between communicator and audiences.

Check Your Progress 2

- 1) All the elements of communication- sender, message, channel, receiver, feedback can be researched. Approaches to research these elements may vary but it is possible to research each and every element of communication process.

- 2) It is possible to design a research study that can examine all the elements of communication process using cultural studies or critical approach.
- 3) Earlier approaches viewed audience as a homogenous entity which was uniformly influenced by the media message. This led to media effects tradition. Later researchers recognised that audiences are diverse and heterogeneous entities who receive media messages and interpret them according to their own context. This also led to plural of audience as 'audiences'. Uses and gratification, Reception analysis and Cultural studies have suggested that audiences are active beings who receive media messages in their local context and do not get uniformly impacted.



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UNIT 3 DEFINING & FORMULATING RESEARCH PROBLEMS

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Selecting the Research Problem
 - 3.2.1 Difference between a Social Problem and a Research Problem
 - 3.2.2 Importance of Review of Literature
 - 3.2.3 Questions of relevance, feasibility and achievability
- 3.3 Formulating the Research Problem
 - 3.3.1 Research Questions
 - 3.3.2 Research Objectives
 - 3.3.3 Research Hypotheses
- 3.4 Defining the Terms of Enquiry
 - 3.4.1 Concepts
 - 3.4.2 Constructs
 - 3.4.3 Variables
 - 3.4.4 Establishing Operational Definitions
- 3.5 Let Us Sum Up
- 3.6 References and Further Readings
- 3.7 Check Your Progress: Possible Answers

3.0 INTRODUCTION

In the previous units, you were introduced to the concept, nature and scope of research. You have read the tenets of scientific research and various classifications and applications of communication research. In this unit, we shall discuss the process of selecting and formulating a research problem. In this regard, conducting survey of available literature is the most important step, though, peers, sponsoring agencies; researcher's own observation of the environment can also suggest a research problem. Once the broad area has been selected, it is then necessary to formulate the research problem in terms of specifics such as research questions, objectives and hypotheses. For this, a review of existing literature in the concerned discipline becomes essential. The more specific and clear you are, the better, since well begun is half done. If you are clear about what you want to research, then how you are going to conduct your research is merely its logical extension.

It is also important to understand the constituents of the terms of enquiry i.e. concepts, constructs and variables. It is their varying values in terms of occurrence, degree of intensity, nature of direction etc. that enable us to draw conclusions about the associations, correlations and causality that exists in human and social phenomena at any given point in time and space.

3.1 LEARNING OUTCOMES

After working through this unit you should be able to:

- differentiate between a social and a research problem;
- identify the factors that determine the selection of a research problem;
- discuss the importance of review of literature;
- define hypotheses and explain its various types;
- discuss concepts, constructs and variables; and
- explain the need and importance of operational definitions in research.

3.2 SELECTING THE RESEARCH PROBLEM

In this section, we will focus on the difference between a social problem and a research problem. We will also explain the importance of review of what has already been studied so that you do not end up ‘rediscovering the wheel’ unless you are interested in replicating previous studies in order to test or strengthen existing theories. We will finish this section by outlining various factors that go towards making the final decision regarding the selection of research problem.

3.2.1 Difference between a Social Problem and a Research Problem

A social problem involves society at large and therefore has a broader framework than a research problem. A research problem has a limited context that primarily involves either academia or industry. Social problems also deal with a range of issues from politics, culture and religion, while research problems are often limited in scope and are well defined.

A social problem has the element of public concern while a research problem can exist at the level of curiosity alone i.e. when a scientist does research just because s/he wants an answer to a question. It can also exist at the level of profit, for example, when a manufacturer wants to identify the optimum market for a new product.

A social problem (riots, global warming, female foeticide etc.) exists irrespective of the researcher. The researcher, on the other hand, formulates a research problem, for instance, s/he could be interested in the role of media in covering communal riots or awareness of female foeticide.

A social problem deals with mandatory (e.g. the legal age of marriage is 18 years) or normative propositions (e.g. Literacy should serve as the foundation of a nation). A research problem deals with propositions that can be proved or disproved through empirical determination. For example, early marriage leads to health problems or literacy leads to social awareness. The validity of both these statements can be tested through research.

A social problem serves as a trigger or a launch pad for the formulation of research problems. Research findings, on the other hand, provide descriptions, explanations, predictions and solutions to social problems. For instance, research could establish that motherhood at a tender age is associated with death of the mother during childbirth or higher risk of infant mortality. Therefore we can say that the findings of research studies enable the framing of mandatory propositions/laws/policies.

A social problem is a one-time problem. It finishes once it is tackled or until it emerges again. A research problem is formulated by the researcher and can be repeated as per his/her requirement. In this regard, experimental research studies set up in laboratories are a case in point.

A social problem causes disruption in the smooth running of the social order while a research study causes no such disruption other than minor intrusions into the respondents' lives.

A social problem can be dealt with by using intuitive thinking, common sense etc. while a research problem is dealt with using scientific thinking based on empirical logic and reason. Moreover, a social problem can arise all of a sudden, necessitating spur of the moment decisions. A researcher on the other hand systematically proceeds to identify, formulate and lay down a procedure to study a research problem and only then starts data collection and analysis.

3.2.2 Importance of Review of Literature

After the identification of the broad area of study and before the finalisation of the research topic, it is important to consider certain relevant factors. This is where the importance of the review of literature comes in. Existing literature available on our chosen research topic is reviewed with the aim to ensure that our topic is relevant, feasible and achievable within a particular time frame, budget and infrastructure.

Review of literature economises efforts and maximises profit. It gives you an idea of what kind of research has already been done, by whom, where, when and how. As an extension, it also tells you which aspects have been largely ignored or which methodologies have not been used - in effect it allows you to identify the research gaps in subject matter, theoretical perspectives, methodologies etc. You can then decide to either replicate previous research and thereby strengthen existing theory or take up new areas and thereby revise existing theory or formulate new explanations for existing phenomena. A review of literature gives you an idea of the practical problems faced by previous researchers as well as take up on their suggestions for further research. Review of literature also gives inputs regarding methodology, terms of inquiry as well as gain an insight into defining the variables you intend to study in your own research.

Literature refers to all information in printed or oral form that is available on your topic of research interest. For review of literature, one can consult: Indices - these contain alphabetically organised listings of research articles under subject, title and author headings as also information regarding from where the article can be sourced, e.g. Social Sciences Citation Index, Humanities Index, etc. Unit 6 deals extensively with this topic.

Bibliographies are akin to indexes other than that they concentrate more on periodical/journal literature. A bibliography can also be limited to just books published in a particular discipline during a specific period.

Compilations of Abstracts can also be consulted. These are typically arranged according to subject matter and author and contain summaries of specific published articles. In that sense they are more beneficial since one can decide to consult the full study on the basis of whether the abstract's relevance to one's

study e.g., Dissertations, Abstracts, International. Journals and Periodicals are other sources of literature. They are publications of a fixed periodicity and discipline which contain full papers, articles, reviews etc.

Books and Encyclopaedi as are also consulted for an exhaustive analysis and overview of one's interest area. Encyclopaedi as can be particularly useful for framing operational definitions.

General and professional/trade publications are other sources. Articles for the layman appearing in newspapers and magazines can provide interesting perspectives on complex problems in communication and media for researchers to pursue. Many trade publications are published in films, advertising, public relations etc. which can provide case studies in respective areas.

The Internet is an immense source of information. The only drawback is that one has to be careful regarding the credibility of the information that one sources online. It is advisable to refer to the websites of recognised journals and research institutions.

3.2.3 Questions of relevance, feasibility and achievability

A good research topic is one that can be related to a theoretical frame of reference. For instance, if you are interested in doing research in advertising, you can refer to the Uses and Gratifications theory, Diffusion of innovations theory etc. or any other theory discussed in detail in the Course on Communication and Media Studies. This will not only root your research in an existing body of theory, but also make it easier to identify potentially fruitful questions. This theoretical base will also enable your research to find its niche in the existing well of knowledge since your findings will either support the existing theory or help revise it. Even if you are doing research in a completely new area, you can still consult theories from related disciplines and develop a better understanding of your subject. For instance, if you are researching social media addiction, you can always draw upon theories from the discipline of psychology and link them to the study of a mass medium. Review of literature of existing research studies helps to determine how relevant your research would be in respect of the existing knowledge - will it fill a gap, will it strengthen it - in other words, is your chosen research problem significant?

One also needs to determine the feasibility of the topic before starting work on it. A topic might prove unsuitable for investigation simply because the question being asked cannot be answered with the facilities and information currently available. For instance, there is no point taking up a topic that requires classified information for the analysis or any other information that cannot be legally obtained. You may not have the necessary budget or infrastructure to use a required technology to investigate the problem. Often, the necessary technology may not even exist. For instance, it is only now that we have technologies like brain mapping to determine how individuals react to message stimuli.

Moreover, a topic does not lend itself to productive research if it requires the collection of data that cannot be observed or measured with validity or reliability. If controlling for error is an important consideration for you then such a topic may not be very feasible. It is equally important to have a very clear idea of one's own capabilities. If one is not a team person then it would not be feasible

to take up a topic that requires large-scale collaboration. Similarly, it is better to take up a qualitative study if one is not comfortable with numbers or statistical analysis.

Before finalising one's research topic, one also needs to address the question of achievability. You might be engaging in research for the fulfillment of the requirements of a degree, such as a dissertation. Quite naturally, you will be working on a deadline if you have got a research grant or a fellowship for a specific period. Therefore, it is important to ensure that your research topic is not too broad and unwieldy. One tip is to write down your proposed title and then attempt to dissect the topic into small questions. For instance, Computer-mediated Communication as a tool for social relations - this working title will immediately ensure that you draw up boundaries and concentrate on this aspect in order to arrive at closure within a specified time period.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List two differences between a Social problem and a Research problem.

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2) List the sources of information for conducting a review of literature.

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3) How can you determine the relevance of your topic?

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3.3 FORMULATING THE RESEARCH PROBLEM

In this section we will focus on how to formulate the terms of enquiry for your research. These primarily constitute research questions, objectives and hypotheses. It is not necessary that all three be included in every research study. For instance, when conducting exploratory research you may just have some research questions and objectives. You may have research objectives and hypotheses in more structured studies that address issues of association, correlation and causality. A study does not necessarily become weaker in the absence of hypotheses. The

importance of formulating the terms of enquiry lies in the fact that it narrows down the research problem to a manageable specificity, lays down the extent and limits, identifies the relationships proposed to be significant - in short all that you hope to achieve through your study.

The important thing to be kept in mind is that there is coordination in the formulation of the terms of enquiry. For instance, if your research question is 'whether demographic variables impact on internet usage', then the corresponding objective should be to collect data pertaining to the demographics of the Internet user as well as his/her Internet usage patterns. It is only when you have this data that you will be in a position to answer your research question. How the question is answered will prove or disprove your research hypothesis that perhaps could be that 'socio economic status and frequency of Internet use is positively correlated'.

3.3.1 Research Questions

Research questions are usually asked in exploratory research when not enough is known about the research area to justify the framing of categorical hypotheses at the outset itself. Since the researcher is working in an area that has only been studied marginally or not at all, s/he may be unsure of the existence of the determining factors or of the interrelationships. At this point, research is a knowledge-gathering exercise rather than verification. So s/he asks a series of questions, gets many answers, looks for patterns and it is only at the end of research that s/he may be in a position to frame hypotheses to be taken up later in a more structured study.

3.3.2 Research Objectives

When a researcher is working in an area for which extensive literature is already available, it acts as a facilitating mechanism for him/her to straightaway frame the research objectives followed by research hypotheses. The latter is in the nature of an answer- a tentative proposition, the validity of which remains to be tested. For instance, s/he can directly list a hypothesis - 'media consumption and need gratification is positively correlated' - without having to first frame it as a question. While framing the objectives, you have to keep in mind that they should be exhaustive and logically sequenced. For instance, an objective to determine media usage patterns should be listed before an objective about studying the reasons for those patterns. Also, it is important that only those objectives are listed that are empirically achievable; i.e., amenable to being observed or measured.

3.3.3 Research Hypotheses

As mentioned earlier, a hypothesis is a tentative generalisation, the validity of which remains to be tested. Hypotheses enable to test existing theories, suggest new ones as well as help to describe social phenomena with a certain degree of certainty. For instance, a hypothesis such as 'the more the exposure to televised violence, the stronger the belief in a violent world', enables you to put George Gerbner's Cultivation theory to the test. Similarly, a hypothesis such as 'communication breakdown between couples is the primary factor for divorce', if proved, enables you to describe the reasons behind divorce rates in the country. The importance of hypotheses in a research study lies in their ability to bring direction, specificity and focus. They help us to control the intervening variables and enable decision- making and policy formulation.

Hypotheses can be classified on the basis of various criteria; on the basis of source, level of abstraction, on the basis of composition and functionality.

On the basis of source:

- a) The researcher can frame a hypothesis on the basis of his/her own perception or intuition regarding a particular phenomenon. For instance, a researcher may belong to a family where the men only discuss issues regarding their work or business while the women only discuss domestic issues. The researcher here draws upon one's personal experiences and hypothesises that 'conversational engagement is associated with identity perception'. These kinds of hypotheses have an important contribution in building up theory and are useful in exploratory research since they bring forth interesting questions.
- b) The researcher can frame a hypothesis on the basis of existing theory. The genesis here lies not in his/her intuition but in the findings of specific research conducted to study a particular phenomenon. For instance, s/he can take up the want-get ratio formulated by Daniel Lerner and seek to replicate it. S/he too may find that this ratio regularly operates in underdeveloped countries while not so much in the developed world. S/he may also identify gaps in existing research and frame a hypothesis accordingly. For instance, one may find loneliness as a factor in media consumption has not been studied, so s/he can frame a hypothesis such as 'social support and media consumption are associated'.

Level of abstraction:

- a) Level of commonplace impression: These kinds of hypotheses are in the nature of an observation about some empirical uniformity in respect of the phenomena under study. The focus here is on the descriptive, for instance, 'male reporters cover hard core reporting.'
- b) Level of conceptualisation: These hypotheses are based on existing theory where a hypothesis has been more or less proved but is open to falsifiability. For instance, the Two-step flow hypothesis is relatively well-established in understanding how information flow takes place in remote areas. Here, information is routed through the local opinion leaders like the *sarpanch* etc. rather than directly through the mass media.
- c) Level of specification: These hypotheses specify the nature of relationship between the variables that constitute the hypothesis. A hypothesis may state that something tends to be accompanied by something else i.e. it states the association between the variables - 'Education is associated with civic awareness. Specifying the degree and direction of the relationship can also state a hypothesis, e.g. 'Increase in leisure time leads to increase in media consumption. A hypothesis can also be stated in terms of a causal relationship between variables - 'Continuous exposure to computer screens causes dry eye syndrome'.

On the basis of composition/complexity:

- a) A simple hypothesis is one that has only one or two variables. For instance, 'Times of India is the largest circulated newspaper in India'. Or 'Age and Internet usage is associated'.

- b. A complex hypothesis is one that has more than two variables. For example, 'increase in age and socio-economic status leads to increase in media consumption'. These hypotheses are more difficult to test because more the variables in a single hypothesis, the more difficult it is to assess their interrelations - quantitatively and theoretically.

On the basis of functionality/testability:

- a) A theoretical hypothesis is stated in terms of abstract concepts and is untestable in any direct sense.
- b) A research hypothesis is derived from the above but is stated in such a way that it can be tested i.e. it is a functional or a testable or a working hypothesis. Most research methodologists recognise only two hypotheses under this category - Research hypothesis and Null hypothesis. The theoretical and the statistical hypothesis are assumed to be understood as part of the research hypothesis. Here the hypothesis is stated in terms of some operations to be performed i.e. construct language is used. For example, 'IQ levels and comprehensive abilities are positively correlated'. To be able to prove or disprove this hypothesis the constructs of IQ and comprehension are measured with the help of test instruments. To determine the IQ any standardised IQ test can be used. To measure comprehension we could construct a test instrument such as a reading passage followed by questions - the respondents' test score will rate their comprehension capability.
- c) A null hypothesis can be deemed to be the reverse of the research hypothesis or its negation. To take up the above example, the null hypothesis would be, 'IQ levels and comprehensive abilities are negatively correlated' or 'IQ levels and comprehensive abilities are independent of one another'. It is the null hypothesis that is actually tested in research. This practice is followed by convention and also because by testing the null hypothesis first, the researcher is declaring his/her objectivity. Since the null hypothesis is a denial of what researcher holds to be true, by according it priority, s/he declares that s/he is giving all the alternatives first chance to be proved.

The concept of null hypothesis is based on the law of probability. The research hypothesis and the null hypothesis are alternative statements. Both have the probability of being true or false, so if one is proven, its alternative, by logic, is considered disproven (though it should be noted that neither the research hypothesis nor the null hypothesis can be absolutely true or absolutely false under any given test of it. That is why we say that the data failed to support the hypothesis or that the data failed to reject the hypothesis). Moreover, it seems easier to prove something false than to prove it is true - one can say that there could be many reasons for a particular effect, but this one is not the cause.

- d) A statistical hypothesis refers to the technical form in which the hypothesis is stated for purposes of a statistical test, i.e. it is stated in a manner that makes it amenable to the application of the relevant statistical tests of significance. The data is quantified, confidence levels are set, tests are applied, values are obtained and checked to determine whether they fall in the area of rejection or acceptance. It is important to note the difference between significance and importance in research methodology. The former is

mathematical while the latter is subjective, based on experience and judgment. A hypothesis may be rejected on the basis of it not clearing a statistical test, set to certain parameters, but that does not mean that it ceases to be important.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) When are research questions asked?

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2) What is the importance of hypotheses?

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3) Define a Null hypothesis.

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3.4 DEFINING THE TERMS OF ENQUIRY

In this section you will study how hypotheses are framed. The criterion of a good hypothesis is that it should be clear and concise regarding what is to be studied and how. In that sense it should also be testable. Whether the hypothesis is merely descriptive or analytical, it will always have one or more variables that have to be observed or measured. In certain instances this task is easy while in other instances procedures have to be put in place so that the observation or the measurement can take place. Since we are dealing with human and social phenomena, we also have to understand that the same phenomena can have multiple meanings depending upon the context of time, situation and person. Therefore, it becomes important to understand how variables should be defined for a specific research study in order to ensure that the findings are understood in the right perspective.

3.4.1 Concepts

Concepts can be referred to as observables i.e. they are terms that have direct empirical referents. A concept is formed by summarising related observations of characteristics that are either directly observable or easily measured. For instance,

income is a concept that can be deduced through observations of a person's lifestyle or directly measured in terms of monthly/annual income. We can say that a concept is a term formed by generalisation from particulars. We can examine the properties of a particular book - size, weight, thickness, paper quality, print, subject matter, ratio of visual content to verbal content etc. and slot it in the genre of text books, reference books, fiction and so on.

3.4.2 Constructs

Constructs can be referred to as non-observables as they cannot be easily related to the phenomena they are intended to represent. They have no direct empirical referent i.e. they are not directly part of our empirical world. For instance, anxiety, violence, faith, intelligence, social prestige, attitudes and so on. You cannot point to something and say this is faith. To be able to observe or measure faith you will have to frame certain parameters or indicators to do so and depending upon how many of those parameters are met, you will be able to state the quantum of faith possessed by a particular person or group. Therefore, we can say that constructs can be defined as higher level abstractions as they are constructed from concepts at a lower level of abstraction. Often researchers construct their own indicators to concretise a phenomenon that is relatively abstract or they combine concepts to explain a construct. In that sense its exact meaning relates only to the context in which it is found. For instance, to measure the construct of social status, a researcher may combine the concepts of income, education and occupation etc.

3.4.3 Variables

A variable is the empirical counterpart of a concept or construct. It is capable of taking on different values, thus in effect a variable is something that varies. Variables are important because they link the empirical world with the theoretical - they are the phenomena and events that can be measured or manipulated in research. A person's age varies in terms of years, income varies in terms of money earned, and intelligence varies in terms of scores achieved on an IQ test and so on. Variables are used by a researcher to measure how they affect other variables. They can also be manipulated to study the corresponding effect of this manipulation on other variables. Researchers try to test a number of associated variables to develop an underlying meaning or relationship among them. After suitable analysis, the most important variables are retained. These are referred to as marker variables and continue to be built upon as research continues in a particular area. Variables can be classified according to the manner of their usage.

- a) Independent variables are those that are systematically varied or manipulated by the researcher. Dependent variables are those that are systematically observed for concomitant variation as a result of the manipulation of the independent variable. For instance, to measure whether the camera angle has any influence on the credibility that a viewer attaches to a news reader, a researcher may shoot the same news bulletin using three different camera angles - top, low and eye level. S/he then shows this news bulletin to three different groups of people and asks them to rate how credible they perceive the news reader to be. S/he may find that the group that was shown the news bulletin where the news reader is looking directly into the camera found the bulletin to be most credible.

In this example, the camera angle is the independent variable and the effect of its variation is measured on the dependent variable i.e. credibility. Since this is an experimental study and there is active manipulation of the independent variable, this variable can also be referred to as the active variable or the manipulated variable or the experimental variable. The dependant variable here can also be referred to as the passive variable or the measured variable. In non-experimental research where there is no active manipulation of variables, different terms can be used - the variable that is used for making predictions or is deemed to be the cause of something e.g. if poverty is present, so will illiteracy, this variable can also be termed as the predictor or the antecedent variable while the variable that is predicted to be result or is assumed to be affected, is called the criterion variable.

- b) When specifying relationships in a hypothesis, it is important to account for or control for the extraneous or the intervening variables. If this is not done then the findings may be erroneous or invalid. For instance, we may have hypothesised that education leads to modernity. We collect data regarding the two variables and prove or disprove the hypothesis accordingly. But it may so happen that even though education does impact on whether a person develops a more modern outlook or not, but no amount of education will greatly change a person's outlook unless his/her family is also a progressive one. But since we have not factored in the variable of family background, we have erroneously held education to be responsible or not responsible for a modern outlook while actually the causal variable had been the family background.

Since, in human and social phenomena many factors combine to create a particular effect, it is important that the researcher either includes all the variables or s/he controls them. With reference to this example, if the researcher feels that family background could be an intervening variable, s/he can control it by only including those respondents in sample who come from similar backgrounds. By doing so, its effect will be cancelled out of the equation since this variable will now be held constant. When such a thing is done, these variables are also referred to as 'control variables'.

Intervening variables can also be understood as those variables without the intervention of which, the relationship between the independent variable and the dependent variable cannot be established. Suppose you want to study the relationship between age and newspaper readership, you assume that older people read the newspapers more. But without the intervening variable of literacy, the relationship between the independent and the dependent variable will not be complete.

3.4.4 Establishing Operational Definitions

It is very important to define the variables under study as failure to do so impacts upon the reliability and validity of the study. In general, any first attempt at a definition is an application of common sense and everyday life experience. For instance, the most conventional approach to defining age would be to define it in terms of years lived beginning from the date of birth. But then as knowledge grows, concepts evolve, meanings change, perspectives multiply, it becomes important to categorically state how one is going to be studying a particular concept for a particular study. For instance, one researcher may want to study

violence purely on physical terms so s/he defines it accordingly. Another researcher wants to study it from the point of view of verbal violence, yet another wants to draw a distinction between animated and real violence, still another wants to study it only from the perspective of the victim and so on. As complexities increase, so does the need to focus on specifics.

When defining the variables under study, two approaches are followed - convention i.e. framing a definition in terms of what is most commonly understood; or framing a definition in terms of the researcher's own understanding or objectives. In the latter case, the researcher has to ensure that the definition remains within the bounds of logic and reason. The conventional definitions are usually the constitutive, nominal or formal definitions which define a word by substituting other words for it. These are usually the dictionary meanings. For instance, status is defined as a person's standing in society; anxiety is defined as nervousness etc.

The stipulated definitions are usually the operational definitions. An operational definition defines a concept in terms of the operations by which it will be represented in a particular study i.e. it specifies the procedures to be followed in expressing or measuring a concept. In a way, operational definitions are quantifications of the nominal definitions. For instance, violence could be defined as any physical act that results in bodily injury to be measured in time units. Anxiety could be defined in terms of the score obtained on a five-item scale or on the basis of observation of the occurrence of certain physical actions such as restless gestures, perspiration, stuttering etc. Basically an operational definition provides an empirical referent to the variable under study.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What is the basic difference between concepts and constructs?

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2) Define a variable.

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3) What is an operational definition?

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3.5 LET US SUM UP

In this unit we discussed how to go about selecting a research problem. We started with differentiating between a social and a research problem. This was followed by highlighting the importance of conducting a review of literature which helps in identification and selection of the research problem. The importance of ensuring that the topic we select should be relevant, feasible and achievable was also dwelt upon.

We further focused on formulating the terms of enquiry - research questions, objectives and hypotheses. The various types of hypotheses were categorised. How the hypothesis is framed using variables was also described. The difference between concepts and constructs was highlighted and the importance of framing clear operational definitions was emphasised.

It is hoped that the above discussion will help you in defining and formulating your own research problem. In the next unit, we shall take the discussion forward and focus on sampling methods.

3.6 REFERENCES AND FURTHER READINGS

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3.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) A social problem can arise all of a sudden while a research problem is formulated. Secondly, a social problem has the element of public concern while a research problem can exist at the level of curiosity alone.
- 2) Review of literature can be conducted by consulting indices, bibliographies, abstracts, journals, periodicals, books, encyclopedias, general interest publications, trade publications and the Internet.
- 3) The relevance of a research topic can be determined by ensuring that it has a theoretical frame of reference as its foundation, thereby either strengthening existing theory or revising it. Its relevance also exists in its significance - it should fill research gaps in subject matter, make methodology more precise, or have implications for framing of policies.

Check Your Progress 2

- 1) Research questions are usually asked in exploratory research when not enough is known about the research area to justify the framing of categorical hypotheses at the outset itself.

- 2) Hypotheses enable us to test existing theories, suggest new ones as well as help us to describe social phenomena with a certain degree of certainty. From the point of view of methodology, hypotheses ensure a more precise study, they impose boundaries, clarify concepts, and enable controlling for intervening variables and testing for statistical significance.
- 3) A null hypothesis is based on the law of probability and is deemed to be the reverse of the research hypothesis or its negation.

Check Your Progress 3

- 1) The basic difference between concepts and constructs is that concepts have a direct empirical referent, which constructs do not have.
- 2) A variable is a concept or a construct that takes on different values, in effect a variable is something that varies.
- 3) An operational definition defines a concept in terms of the operations by which it will be represented in a particular study i.e. it specifies the procedures to be followed in expressing or measuring a concept.



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UNIT 4 SAMPLING METHODS

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 Population
 - 4.2.1 Sample
 - 4.2.2 Types of sampling
 - 4.2.3 Sampling Error
- 4.3 Non-Probability Samples
 - 4.3.1 Accidental Samples
 - 4.3.2 Available/Convenience Samples
 - 4.3.3 Volunteer Samples
 - 4.3.4 Purposive Samples
 - 4.3.5 Quota Samples
 - 4.3.6 Snowball Sample
- 4.4 Probability Samples
 - 4.4.1 Simple Random Sampling
 - 4.4.2 Table of Random Numbers
 - 4.4.3 Systematic Random Sampling
 - 4.4.4 Stratified Random Sampling
 - 4.4.5 Cluster Sampling
 - 4.4.6 Multi-Stage Sampling
- 4.5 Sampling Size
- 4.6 Let Us Sum Up
- 4.7 Keywords
- 4.8 References and Further Readings
- 4.9 Check Your Progress: Possible Answers

4.0 INTRODUCTION

You use a small spoonful of curry to taste what the whole dish is like. Based upon what you find, you choose to eat or not to eat the curry. That small spoonful is a sample, a reflection of the whole dish, not too different from what is done in research designs.

In statistics and survey methodology, sampling is concerned with a subset of individuals from within a statistical population to estimate the characteristic of a whole population. The whole is called 'population, which could be any set of definable elements. That group or sample can stand for or represent the whole. An element may be persons, households, organisations, television programmes or a spoonful of rice or stew. Sampling involves the selection of elements to represent the population. So in other words, a fully representative sample has characteristics similar to the population in the same proportion to the population. Populations are defined according to the purpose of research. Therefore, the first step in sampling is to project the intention of the research.

The selection of elements in a sample is made from a sampling frame which identifies all the individual sampling units within a population. A sampling unit may be a single element or a collection of elements. Examples of sampling unit could be a household, a television programme or a child. Typical sampling frames can be telephone directories, voter registration lists, PIN code lists, list of licensed television channels, memos distributed in an organisation and so on.

4.1 LEARNING OUTCOMES

After working through this unit, you should be able to:

- discuss the meaning of population in research;
- identify the types of sampling methods used in research;
- distinguish between probability and non-probability sampling;
- adopt a sampling design with minimum error;
- apply the various sampling techniques in research; and
- generalise the results of the sample to the population.

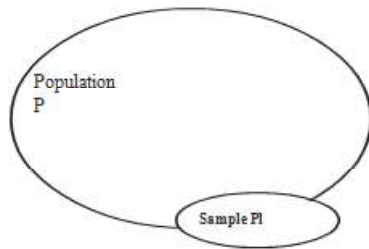
4.2 POPULATION

All items under consideration in any field of enquiry constitute a 'universe' or 'population'. The goal of any research is to first describe the nature of a **population**. It could be a-group or class of subjects, variables, concepts or phenomena. Sometimes an entire group needs to be investigated. The process of examining every member of such a population is called a census. For instance, India's census taken every ten years enumerates the entire population. Examining an entire population is not possible in practice under many circumstances. It involves a great deal of time and resources. Also, studying every member of a population can confound the research because measurements of large numbers often affect measuring quality. Hence, we select only a few items from the universe for the purpose of our study. The items so selected constitute what is technically called a sample and the procedure adopted to select that sample is the sampling technique or sample design.

Populations are defined according to the purposes of research. If the purpose of research is to predict the outcome of the elections in a state, then the population is composed of the actual voters in that election. If we wish to predict the outcome before the official counting but after the vote have been cast, then we can do an exit poll of voters as they leave their voting booths. But, if our intent is to predict the outcome before the day of election, then we have to redefine our population. Here, our population would comprise of registered voters who may be further qualified with a firm intention to vote.

4.2.1 Sample

A sample is a subset of the population that is representative of the entire population. In the diagram below, P is the population and P1 is the sample. A census would measure every member of the population i.e., P whereas a sample would be a portion of the population i.e., P1. The most important part of any sampling procedure is to avoid any kind of bias i.e., every element or member of the population should have an equal chance of being selected. In other words, the sampling design must be free from any bias.



4.2.2 Types of Sampling

Sampling is of two types: probability and non-probability.

Probability and Non-Probability Samples

Samples can be either probability or non-probability samples. A probability sample is selected according to mathematical guidelines where the chance of selection of each unit is known. A non-probability sample does not follow the guidelines of mathematical probability. The distinguishing characteristic of both these types of samples is that probability samples allow researchers to calculate the sampling error whereas non probability samples do not. Probability samples also enable a more accurate generalisation of the results of a survey.

4.2.3 Sampling Error

There is always a need to make a sample error free. However, often, this may not be possible. The degree to which measurements of the units or subjects selected differ from those of the population as a whole is known as sampling error. The procedure adopted to select the sample introduces the likelihood of sampling error. Sampling error is also known as standard error. In other words, sampling error occurs when measurements taken from a sample do not correspond to what exists in the population. Therefore, while selecting a sampling procedure, the researcher must ensure that the sample design causes a relatively small sampling error and helps to control the bias.

To minimise sampling error, the characteristics of a good sample design should adhere to the following criteria:

- 1) The sample design must result in a truly representative sample.
- 2) The sample design selected must ensure minimum sampling error.
- 3) The sample design selected should take budget and time constraints into consideration.

Sampling error does not depend on the type of sample. Whether convenience, judgment or probability sampling, the possibility of error always exists when population members vary and something less than the entire population is drawn. With convenience and judgment samples, one can do nothing more than accept that possibility. With probability samples, we can estimate the rate of sampling error, thereby giving us an idea of the risk assumed in accepting the sample as true.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

- 2) Compare your answers with those given at the end of this Unit.

Fill in the blanks:

- 1) The process of sampling involves the selection of a group of individuals from a
- 2) The selection of elements in a sample is done within a
- 3) occurs when the sample selected does not represent the population.
- 4) A sample is selected according to mathematical guidelines.
- 5) A sample design selected should take and into consideration

4.3 NON-PROBABILITY SAMPLES

There are other issues to be taken into consideration while deciding to use probability or non-probability samples. Some research studies are not designed to be generalised to the population but to collect exploratory data for designing questionnaires or measurement instruments. A non-probability sample is appropriate in such situations. Secondly, if the cost of a probability sample is too high in relation to the type and quality of information collected, then a non-probability sample is a possible alternative. Since, probability samples are often time consuming, a non-probability sample may be adopted to meet the time-constraints.

Sometimes the sampling frame may not contain accurate information. For instance, if a telephone survey is used in elections to predict voting patterns, and the telephone directory is used, those who do not have access to telephones are left out. Sometimes, especially in developing countries, data enabling the creation of a sampling frame simply does not exist.

Although non-probability sampling may be appropriate in certain situations, it is always best to use a probability sample when a study is conducted to support or refute a significant research question or hypothesis and the results will be generalised to the population. We take up these two designs separately:

In non-probability sampling, the items for the sample are selected deliberately by the researcher. In other words, the researcher purposively chooses particular units of the universe for constituting a sample. Mass media researchers frequently use non-probability sampling, particularly in the form of available samples, samples using volunteer subjects and purposive samples. For example, in testing drugs, volunteer subjects have to be used, given ethical and other considerations. Some of the different types of non-probability samples are:

- a) Accidental samples
- b) Available/Convenience samples
- c) Volunteer samples
- d) Purposive samples
- e) Quota samples
- f) Snowball samples

4.3.1 Accidental Samples

In accidental sampling, the researcher simply reaches out and selects the subjects that s/he comes across and continues doing so till such time as the sample reaches a designated size. For example, s/he may take the first 150 persons one meets at a mall entry point who are willing to be interviewed or to provide the information s/he is seeking. In such a sample, there is no way of estimating bias except by doing a parallel study with a probability sample or undertaking a complete census. This does not mean that accidental samples have no place in scientific research. Besides being economical and convenient, they can provide a basis for stimulating insights and hypotheses.

4.3.2 Available/Convenience Samples

An available sample is also known as a convenience sample. It is a collection of readily accessible subjects for study, such as a group of students enrolled in a mass media course or shopkeepers in a mall. Although available samples are helpful in collecting exploratory information, the samples may contain unknown quantities of error. Researchers need to consider both the positive and negative aspects of available samples before using them in a research study. Available samples are a subject of debate in research as critics argue that these samples do not represent the population and therefore have no external validity. Proponents of available samples claim that if a particular trait or characteristic does exist, then it should exist in any sample. This sampling technique is useful in pretesting questionnaires or conducting a pilot study.

4.3.3 Volunteer Samples

Persons who willingly participate in research projects are known as volunteer samples. Subjects who constitute a volunteer sample also form a non-probability sample as the individuals are not selected according to mathematical guidelines. Researchers have found that these subjects tend to exhibit higher educational levels, occupational status and intelligence levels. These characteristics imply that the use of volunteer subjects may significantly bias the results of the research and may lead to inaccurate assumptions of various population parameters. In some cases, volunteer subjects are necessary but they should be used carefully since they contain unknown quantity of error. Volunteer samples are extensively used these days by the media and internet websites. Various polls conducted on radio and television stations, TV networks, the Internet, newspapers and magazines use volunteer samples. However, such samples are considered inappropriate in scientific research.

4.3.4 Purposive Samples

The basic assumption behind purposive sampling is that the subjects are selected for a specific characteristic or quality and eliminates those who fail to meet these criteria. Purposive samples are often used in advertising studies where researchers select subjects who use a particular type of product and ask them to compare it with a new product. However, in such a sampling there is no assurance that every element or subject has some specifiable chance of being selected. Here, the sampling errors and biases cannot be computed since the sampling procedure does not involve probability sampling at any stage.

4.3.5 Quota Samples

One of the most commonly used methods of sampling in market research is the method of quota sampling. Here the subjects are selected to meet a predetermined or known percentage. The basic objective of such sampling is the selection of a sample that is similar to the population in terms of proportion of certain characteristics. For example, a researcher is interested in finding out how girl students differ from boys in their intelligence levels in a co-educational institution. If, there is a sharp difference in the proportion of girls and boys studying in the institution, then, a quota sample is appropriate in order to reflect the population characteristics. For this, the researcher would identify the proportion of the subgroups (girls and boys) in the population (the institute) and apply the same proportion while selecting the sample. So, if there are 200 girls and 100 boys in the institute, the selected sample would have double the number of girls as compared to the boys. In quota sampling the population is reflected in terms of certain characteristics, and the proportion of the population with specific characteristics is determined and selected like-wise.

4.3.6 Snowball Samples

Sometimes, especially in exploratory research, it is difficult to find the relevant respondents. Therefore, each respondent is asked to identify one or more respondents, e.g. people with walking sticks; cancer specialists; experts in a particular field, or mothers who also volunteer their services, even other friends who are social media users. Thus, the sample grows to a predetermined number by each respondent identifying others who can be sampled.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

Fill in the blanks:

- 1) An available sample is also known as a sample.
- 2) Persons who willingly participate in a research project are known as samples.
- 3) Insampling persons are selected to meet a predetermined percentage.
- 4) The basic assumption behind sampling is that the subjects are selected for a specific quality.
- 5) In sampling, the researcher simply reaches out and selects the subjects that s/he comes across.

4.4 PROBABILITY SAMPLES

Probability sampling method involves a systematic selection procedure to ensure that each unit has an equal chance of being selected. However, it does not always guarantee a representative sample from the population, even when systematic selection is followed. It is possible to randomly select 50 students of a university hostel in order to determine the average number of hours spent on watching

television during a typical week only to discover that there was no TV set installed in the hostel or even if it was installed it was never in a working condition. This may be unlikely but it underscores the possibility of inadequacy.

The most commonly used probability samples are:

- 1) Simple random samples
- 2) Systematic random samples
- 3) Stratified random samples
- 4) Cluster samples

4.4.1 Simple Random Sampling

The most basic type of probability sampling is the simple random sampling. Here, each subject or unit in the population has an equal chance of being selected. In principle, one can use this method for selecting random samples from populations of any size. But in practice, it becomes very cumbersome.

If a subject or unit is drawn from a population and removed from subsequent selections, the procedure is known as random sampling *without replacement*- a widely used random sampling method. Random sampling *with replacement* involves returning the subject or unit to the population so that it has an equal chance of being selected another time.

4.4.2 Table of Random Numbers

Researchers also use the list of random numbers to generate a simple random sample. For example, a researcher wants to analyse the portrayal of women in 10 soap operas on television channels out of a population of 100 programmes then s/he can use the table of random numbers to select 10 programmes by numbering each of the 100 programmes from 00 to 99. A random number table is included as an Appendix. First a starting point in the table is selected. There is no specific way to choose a starting point; it is at the discretion of the researcher. The researcher then selects the remaining 9 numbers by going left, right, up or down. For example, if the researcher goes down the table from the starting point 39 then his drawn sample will include programmes numbered 39, 02, 78, 94, 71, 83, 20, 49, 64, 08 and 55.

Simple random samples for use in television surveys are often obtained by a process called random digit dialling. This method involves the randomly selected four-digit numbers and adding them to the three-digit or four-digit exchange prefixes in the city in which the survey is conducted. Many of the telephone numbers generated by this method are invalid because some phone numbers are disconnected or they may be temporarily out of service and so on. Therefore it is best to consider three times the number of telephone numbers needed; if a sample of 100 is required then at least 300 telephone numbers should be generated.

Random number generation is possible through a variety of methods. However, two basic rules must be kept in mind: (1) each subject in the population must have an equal chance of being selected (2) The selection process must be free from bias of the researcher. The purpose of random sampling is to reduce sampling error and overlooking the above mentioned rules only increases the chance of error creeping into the study.

4.4.3 Systematic Random Sampling

The most practical way of sampling is to select every 1st item on a list. Sampling of this type is known as systematic random sampling. For example, to obtain a sample of 50 from a population of 500, or a sampling rate of 1/10, a researcher randomly selects a starting point and sampling interval. Thus, if the number 4 is chosen as the starting point, then the sample will include the 50 subjects or items numbered 4, 14, 24, 34, 44, 54, and so on. To add further randomness to the process, the researcher can select the starting point and the sampling interval randomly. For example, an interval of 6 with a starting point 37 generates the numbers 43, 49, 55, 61, and so on.

Let us take an example. You wish to study the contents of a newspaper over a three month period with a sample of 30 issues. If you take a simple random sample, it is possible that all the days of the week may not be included. So you choose a systematic random sample choosing every 3rd issue. This way, the likelihood of all days of the week being included is ensured. In systematic random sampling, selection of subjects is easy, accurate and generally inexpensive as compared to simple random sampling. However, a complete list of the population must be obtained as hidden periodicity in the population may bias the process.

4.4.4 Stratified Random Sampling

If a population from which a sample is to be drawn does not constitute a homogeneous group, stratified sampling design is generally used in order to obtain a representative sample. In some cases, the researcher wants to ensure that a specific sub sample gets adequate representation. In such instances, a stratified random sampling approach is adopted to get an adequate representation of the sub sample. The population is divided into different strata with specific characteristics. The characteristics or features of the strata or segment may include any variable: age, gender, religion, occupation, income level etc. There are some benefits of stratification: 1) Sampling from strata ensures that at least some elements of the total sample will come from each strata 2) Since the procedure involves choosing independent samples from each strata, different methods of selection can be used for one or more strata 3) Sampling error is considerably reduced because each strata has an assured representation in the sample.

Stratified random sampling can be applied in two different ways. Proportionate stratified sampling includes strata with sizes based on their proportion in the population. If 20% of the population is in their teens in the age-group of 13-19, then 20% of the total sample will be subjects in this age-group. This gives each person in the population an equal chance of being selected. In disproportionate stratified sampling, the sample units are selected on equal basis from each stratum irrespective of their size of representation of their strata in the population.

Some merits of the stratified random sampling method are that selection of the sample units is made from a homogeneous group and comparisons can be made to other populations. One of the disadvantages of this method is that this method is time-consuming as well as expensive.

4.4.5 Cluster Sampling

A key difference between cluster sampling and stratified random sampling is that the total population is divided into clusters or groups or categories and the

sample units are randomly selected from each of these clusters. The ultimate sample consists of all units from these clusters. With cluster sampling, the researcher can divide the state into districts or PIN code areas and select groups from these areas. For example, a PIN code area may contain mostly residents of a high socio-economic status who are unrepresentative of the rest of the state; if selected for analysis, such a group may confound the results. To help control such error, it is best to use small areas or clusters, both to decrease the number of elements in each cluster and to maximise the number of clusters selected (Babbie, 2001). If clusters are formed or based on some geographic sub-divisions, then cluster sampling is better known as area sampling.

4.4.6 Multi-Stage Sampling

Multi-stage sampling is a further development of the principle of cluster sampling. In many nation-wide surveys, researchers use a form of cluster sampling called multi-stage sampling, in which individual households or persons and not groups are selected. In multi-stage sampling, each stage is a separate sampling task. Therefore, the separate stages can be approached with different methods. For example, in a large population, stratification may be useful within the initial steps and simple random procedures in final ones.

Let us take an example of a city, where one does not have exact data of all the residents. You can divide the city into its zones from available municipal data, divide the zones further into wards, and select clusters from wards, ensuring that all zones are represented. So, if there are 6 zones and 10 wards in each zone (total 60), you may choose two wards from each zone, and then choose two clusters (each of 10 persons) from each ward. In this way, you can get an adequate sample of 120 respondents, representative of the whole city.

Check Your Progress: 3

- Note:** 1) Use the space below for your answers.
2) Compare your answers with those given at the end of this Unit.

Fill in the blanks:

- 1) The purpose of random sampling is tosampling error.
- 2) Simple random samples for use in television surveys are often obtained by a process called
- 3) If a subject or unit is drawn from a population and removed from subsequent selections, the procedure is known as
- 4) In sampling, population is divided into different and specific characteristics.
- 5) If are formed or based on some geographic sub-divisions, then cluster sampling is better known as area sampling.

4.5 SAMPLING SIZE

Every sampling procedure has to deal with the question of how large or small a sample to draw. Unfortunately, this simple question does not have a simple answer. Ideally, larger samples are preferable, but this is often not possible. The size of

the population is dependent on certain considerations of the project under study. 1) The resources and time available 2) the complexity of the characteristics of the population under study 3) the precision required to approach the study. The important aspect to understand is that the sample, however selected, should be as representative of the population as possible, whether very small or very large.

The complexity of a characteristic depends on the number of categories used to measure it. Say for example, the likelihood of a respondent watching a reality show on television. One can begin asking “Yes/No” question. Then we may investigate the “Yes/No” responses by the sex of the respondent, 3 levels of socio-economic class and 6 levels of education and so on. In this case we can have a 2x2x3x6 category scheme or 72 different cells.

Precision is determined by the efficiency of the statistical procedure to be used and the variability of the characteristic under study. It does not involve the size of the population. Here, the researcher’s time and money are important.

Characteristics which are complex and must be approached with high precision require large samples and considerable resources. A rough rule of thumb for the novice researcher is that samples of less than 30 are generally considered inadequate; samples in the 100 to 200 range are rarely brought into question on the basis of size and some research questions require samples of greater than 500.

Sampling Bias

Sampling bias refers to a sampling procedure which systematically excludes some member or members of the population. Whereas convenience and judgment samples are by definition biased samples, true random samples are free from sampling bias but not sampling error by their nature. Most probability samples are not truly random because of the difficulty of effecting random selection. Sampling bias is also the consequence of respondents refusing to participate in a study even when drawn in a sample.

Check Your Progress 4

Note: 1) Use the space below for you answers.

2) Compare your answers with those given at the end of this unit.

Fill in the blanks:

- 1) refers to a sampling procedure which systematically excludes some member or members of the population.
- 2) is determined by the efficiency of the statistical procedure to be used and the variability of the characteristic under study.
- 3) What are the considerations taken into account while determining the size of a population for a research study?

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4.6 LET US SUM UP

In this unit we have discussed the methods employed to select a sample from a given population. To make predictions about events, concepts or phenomena, researchers must perform detailed objective analyses. One procedure to use in such analyses is a census, in which every member of the population is studied. However, conducting census for each research project is impractical. The most widely used option is to select a random sample from the population and make predictions from it that can be generalised to the population.

If the scientific procedure is to provide valid and useful results, close attention must be paid to the methods used in selecting a sample. Probability and non-probability samples are adopted depending on the kind of research, costs and time. The non-probability samples are accidental, available, volunteer, purposive and quota samples while the probability methods are simple random sampling, systematic random sampling, stratified random sampling and cluster sampling. The size of the sample required depends on many factors - the purpose of research, time and financial constraints. To sum it up, it is important for any researcher to seriously consider the various factors discussed above to select a research sample for any study.

4.7 KEY WORDS

Cluster sample: sample placed into groups or categories.

Population: a group or class of objects, subjects or units

Sample: a subgroup or subset of a population or universe.

Sampling interval: a random interval used for selecting subjects or units in the systematic sampling method

Sampling frame: a list of members of a particular population

Sampling error: the degree to which measurements obtained from a sample differ from the measurements that would be obtained from the population

4.8 REFERENCES AND FURTHER READINGS

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Wimmer and Dominick (2003), *Mass Media Research-An Introduction*, Thomson Wadsworth, Singapore

Wilkinson and Bhandarkar (1990), in *Methodology and Techniques of Social Research*, Himalaya Publishing House, Delhi

4.9 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) Population 2) Sampling Frame 3) Sampling Error 4) Probability Samples
- 5) Budget and Time

Check Your Progress: 2

- 1) Convenience Sample 2) Volunteer Samples 3) Quota Samples 4) Purposive Samples
- 5) Accidental Sampling

Check Your Progress: 3

- 1) Reduce 2) Random digital dialing 3) Random sampling without replacement
- 4) Random stratified, strata 5) Clusters

Check Your Progress: 4

- 1) Sampling bias
- 2) Precision
- 3) The size of the population is dependent on certain considerations of the project under study.
 - 1) The resources and time available
 - 2) The complexity of the characteristics of the population under study
 - 3) The precision required to approach the study.

UNIT 5 REVIEW OF LITERATURE

Structure

- 5.0 Introduction
- 5.1 Learning Outcomes
- 5.2 Literature Review: Need and Importance
 - 5.2.1 Literature Review for Whom?
 - 5.2.2 Focus of Review of Literature
- 5.3 Objectives of Review of Literature
 - 5.3.1 Refine ideas and concepts
 - 5.3.2 Sharpen research objectives and hypothesis
 - 5.3.3 Select appropriate theoretical framework and research methods
 - 5.3.4 Identify practical problems
 - 5.3.5 Avoid duplication of work and generate ideas for future research
- 5.4 Evaluation of Material for Review
 - 5.4.1 Selection of Material
 - 5.4.2 Collection of Material
- 5.5 Writing Review of Literature
 - 5.5.1 Organisation of the Material
 - 5.5.2 Writing the Reviews
 - 5.5.3 Ethical issues
- 5.6 Let Us Sum up
- 5.7 References and Further Readings
- 5.8 Check Your Progress: Possible Answers

5.0 INTRODUCTION

Review of Literature is an important component of research work. Whether you are working on your research paper, dissertation or thesis, literature review is an integral part of the research process. However, many a time it is not properly understood and carried out. Students/ research scholars often consider review as a collation of large body of information and statistics available on the subject.

The literature review has a specific purpose and role in research - it collates, classifies and summarises the major work undertaken in the research area. Review of Literature traces the existing scholarship available in a specific field, their focal areas, research approaches and methodologies used and major findings arrived at. Thus, it captures the essence of the studies in a specific area and links them with the present research work.

Review is concerned with secondary data and discusses published studies. It draws on the heritage of the past by synthesising the ideas and arguments made by other scholars. It captures major debates in a specific area and provides new interpretations to the existing available material. Thus it serves as a handy guide of intellectual progression in an area of study.

We shall discuss various aspects of Review of Literature in detail in this unit. We shall start with the need and importance of literature review in research work,

the process of selection and organisation of material collected and how to write a review.

5.1 LEARNING OUTCOMES

After reading through this unit, you should be able to:

- discuss the need and importance of Review of Literature in research;
- describe the objectives of Review of Literature;
- outline the steps in material selection and collection for literature review; and
- describe the process of writing Review of Literature.

5.2 LITERATURE REVIEW: NEED AND IMPORTANCE

In Unit 4, subsection 1.3.2 while discussing the characteristics of research, we had discussed that scientific research is:

- Objective: as it deals with facts;
- Empirical: as it deals with the knowable and potentially measurable facts/elements;
- Predictive: as it relates the present to future and predicts behaviour;
- Systematic and cumulative: the studies previously undertaken serve as building blocks; and
- Public: as the results are shared with scientific community to build knowledge.

For the purpose of our discussion, let us focus on the last two points - scientific research is **cumulative** and **public**. Research, as you are aware, is a series of work undertaken by different scholars in an area of exploration. They have examined varied aspects of an issue over a period of time thus added a body of scholarship in a discipline and also shared their work with larger public. This sharing has been done through various means such as publications in research journals, books, theses and dissertations as well as presentations in seminars and conferences, and so on. Thus, a large body of research material already exists in a subject that needs to be accessed, examined and referred to. It is possible that the available material may not directly be related to the specific area which you wish to investigate. However, it may be similar in some respects or have linkages with your topic, hence merits a close examination or '**review**'. Thus review of literature enables you to examine the gamut of issues and available research involved in an area, create a broad picture and situate your own work within a larger framework.

As discussed in Unit 2, primary data and secondary data are used in research. Primary data collection involves first hand observation by researcher while secondary data is already available - possibly for reasons other than the purposes of your research. The Literature Review is concerned with secondary data in which major research findings of other scholars and researchers in specific as well as related areas are collated, classified and summarised.

5.2.1 Literature Review for Whom?

When you start working at the review section you need to ask the question, for whom I am writing the literature review? This will help you select appropriate material and pitch it at the proper level. The literature review is certainly not meant for a general audience - the audience/ readers of review of literature will be specialised in nature. For example, at the Masters level, they will be department faculty, dissertation supervisor, and academic councilors at the Regional Centres among others. Similarly, at the PhD level, they will be your supervisor /s, fellow researchers, Doctoral Research Committee (DRC) members and examiners. Scholars in other disciplines, media practitioners and even policy makers may also read your thesis. Thus, at each level, your work will be critically examined by a specialised audience emphasising the need for suitably pitching the content.

5.2.2 Focus of Review of Literature

One question often asked by researchers is that what type of information needs to be cited in Review of Literature and how is it different from the information given in the introductory chapters. To this question, it is suggested that while accessing material, you will come across a vast body of literature in your subject that will include general information, statistical data, opinions and observations of scholars, and research studies. All information may not be useful for your research work, and you will have to select relevant information with care.

You may include the theoretical content, statistical data etc. in your introductory chapters to develop and build a context for your study. Ideally, the focus of review of literature should be on research studies. Therefore, while undertaking library/ Internet research, you should give special attention to the studies undertaken in your area for inclusion in literature review section, apart from collecting background material.

You may also include studies in related fields to develop a broader perspective on a subject. For example, if you are investigating Health Communication for Polio - apart from polio, you may also collect studies undertaken on communication for sanitation, hygiene, nutrition, reproductive health and so on. This will enable you to examine the common patterns in other aspects of health, to provide a holistic view. However, if a good number of studies are available in your specific topic, you may limit the studies to polio, to keep the discussion focused.

5.3 OBJECTIVES OF REVIEW OF LITERATURE

According to Berger (2000) the literature search does two things: 'first it offers a sense of context for your readers so that they can see how your research fits into the larger scheme of things. Second, it shows readers where you got your information and lets them assess how current it is and how reliable it might be'. Thus the information collected for literature review has to be relevant, current, and reliable.

In addition, the review also helps to:

- refine ideas and concepts
- sharpen research objectives and hypothesis

- select appropriate theoretical frameworks and research methods
- identify practical problems
- avoid duplication of work and generate ideas for future research.

5.3.1 Refine Ideas and Concepts

The review of literature helps in firming up your ideas and developing a sound grasp of your topic. When you venture into research, you have great interest in a specific subject but may not know much in depth about it. The rich resources expose you to the varied research areas/ problems undertaken and thus enable you to evaluate your concepts and approaches and in turn firm them up. According to Brennen (2013), “literature review should not only give readers an understanding of what research has been done in this area but also describe how the new research project fits into the broader field of study and why it is important to pursue the study”.

5.3.2 Sharpen Research Objectives and Hypothesis

Designing proper objectives is a skill which needs to be developed by researchers. A study of available literature helps you to design and sharpen your objectives properly. Many a time, objectives are written in general and vague terms such as ‘to know’, ‘to find out’, ‘to understand’ etc. which are not observable and measurable. A good research study outlines the research objectives in behavioural terms which are measurable and specific. Some examples of behavioural terms are given in the box below:

Behavioural Terms

Analyse	Examine	Explore	Explain	Identify
Illustrate	Outline	Define	Describe	Distinguish
Demonstrate	Differentiate			

Apart from refining your ideas and research objectives, the findings of other studies help you in formulating your hypothesis which is an important element of the research design. Hypothesis, as discussed in Unit 3, is a tentative proposition, the validity of which remains to be tested. A proper hypothesis ensures a more precise study, as it imposes boundaries and helps to control intervening variables and testing for statistical significance. The review of literature exposes you to different forms of writing hypothesis and enables you to write it in a clear and concise manner.

5.3.3 Select Appropriate Theoretical Framework and Research Methods

The review of literature helps in selection of appropriate theoretical framework and research methods. As discussed in previous units, theoretical framework links a study with the larger body of work undertaken in a specific area and thus helps to focus and guide your inquiry.

Similarly, you can learn from the strong points of previous research - the type of research approach used (qualitative; quantitative or mixed method), research methods, tools and techniques selected and so on. A critical assessment of the

research methodology adopted by other scholars will help you select suitable tools and techniques for your research objectives. However, care needs to be taken if you are making a major deviation, for example, if majority of the studies reviewed have adopted qualitative approach and you wish to use quantitative methods, you will have to give a proper justification for such a deviation.

Some studies may not use proper research framework or sample selection hence could be weak in design, treatment and analysis. There could be some contradictions and inconsistencies in the results which may not be valid and reliable. Thus reviewing a large body of literature will bring criticality and the learning acquired will enable you to modify and select appropriate research tools, and thus infuse rigour in your research design.

5.3.4 Identify Practical Problems

Many reports mention some practical problems faced by the researchers - these could be in terms of vast scope of the study (taking up too many objectives or research methods, large sample etc.) which they found difficult to achieve within the limitations of timeframe, efforts and resources. The problems could also relate to some field experiences which were avoidable. As a researcher you can learn from the mistakes of others and avoid the pitfalls in your own study. Thus literature review helps to delimit the scope of the study and economise your efforts.

5.3.5 Avoid Duplication of Work and Generate Ideas for Future Research

A critical review of existing materials helps to determine which types of research, objectives, themes and topics have been 'over emphasised, under emphasised and/ or neglected and what more can be done to make the material more comprehensive'. According to Priest (2010), 'a review of literature helps clarify conceptual issues, identify fruitful theories and methods and rule out "dead ends" that appear to have been thoroughly investigated already, which is why it is first on the list of essential steps in the research process'. Many published research reports generally include suggestions for further research in an area which are helpful in generating ideas for future research.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What is the role of theoretical framework in a study?

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2) Why research objectives should be written in behavioural terms?

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5.4 EVALUATION OF MATERIAL FOR REVIEW

The library is the first place to initiate the process of selection and collection of material and the available resources help you gain some familiarity with your field of research.

5.4.1 Selection of Material

Evaluation of material starts with the gate-keeping process itself. At the first level; credibility of the source needs to be ensured, for example, in the case of a journal, you need to find out whether it is peer-reviewed or not, the periodicity and regularity of publication etc. The authority of the writer, currency and accuracy of content etc. also need to be evaluated. An assessment of the definition of concepts used and the evidence provided also help in ensuring quality of the material selected.

During library as well as Internet research you will come across a variety of information from varied sources of superficial nature. However, the kind of information needed for your specific purpose needs to be ensured otherwise you will get saddled with a large body of material. While some amount of extra material is inevitable during collecting material for review, care needs to be taken to remain focused on the research topic. It is also possible that in your anxiety to include large amount of information the depth of information gets ignored. As a researcher you should ensure that the information selected is very specific and in-depth to make your review of literature chapter a meaningful exercise.

A key point to remember is that a source like Wikipedia is not acceptable as an academic source for research.

5.4.2 Collection of Material

Libraries mostly use a system of keywords in their indexes which should be used to identify the relevant material. Material can also be located by examining the citations in recent publications on your topic. Search engines are commonly used for identifying relevant literature but much of the material may not be reliable or relevant for your work. Newspaper articles also give latest data and statistics on some areas from time to time. Academic journals constitute the most reliable and credible sources for literature review followed by books and monographs. The following are some well-known peer-reviewed journals in mass communication:

Journals in Mass Communication

- Journal of Creative Communications, Sage Publication with MICA, Ahmedabad
- Vidura, Press Institute of India, Quarterly, Chennai
- Communicator/ Sanchar Madhyam, IIMC, Quarterly, New Delhi
- Media Asia, Journal of Asian Media Information and Communication Centre (AMIC), adopted by Taylor and Francis, Singapore
- The Asian Journal of Communication, Wee Kim Wee School of Communication and Information jointly with Asian Media Information and Communication Centre (AMIC), Singapore

- Communication, Culture & Critique - International Communication Association (quarterly), Southern Illinois College of Mass Communication, USA
- Communication Research Trends (Quarterly), Communication Department, Santa Clara University, Santa Clara, USA
- Journalism & Mass Communication (Quarterly), Sage Publication, School of Journalism and Mass Communication, University of North Carolina, Chapel Hill, USA
- Media, Culture & Society - Sage Publication (quarterly), University of Glasgow, UK

Since research process, especially academic research involves a prolonged period of study; it is advisable to keep records of the data accessed which should be done concurrently, lest you may forget. The downloaded material from online resources may be kept in separate folders in your computer with appropriate titles/ captions. Apart from ensuring reliability and credibility of online resources, you must note down the date of retrieval for the purpose of citation.

Relevant pages from books and journals may be photocopied and kept in separate file covers. There will, however, be occasions when you will have to take copious notes of the material available in an exclusive library not quipped with photocopying facility. With the increased accessibility of smartphones fitted with cameras, researchers take pictures of relevant pages and convert them into .pdf files. However, some libraries/organizations/individuals may not allow you to take pictures without permission. Hence, it would be prudent to take prior permission of the concerned, in view of the ethical issues involved.

Whatever strategy is followed for material collection, you must ensure full bibliographic details of all the resources collected so that you do not lose out an important citation at the report writing stage due to misspelling of a foreign author's name or missing year of publication, and so on.

Yet another common query is that how many studies or citations are considered sufficient for a review. In this regard, there is no rule of thumb and it would entirely flow from your research topic, availability of the material as well as the level of study (Masters, MPhil, PhD). Priest (2010) argues, "you will keep seeing citations to literature that has already been examined as key academic sources on a specific question are usually limited in number and often refer to one another".

Once requisite material has been collected, it has to be thoroughly read, re-read, classified and arranged for writing.

Activity-1

Identify a topic of your choice for exploration. Select a research journal in media and communication and obtain the issues of past three years. Go through these issues and find out if similar or related topics were investigated by other researchers in the journal. List the studies identified.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What care will you take while selecting material for literature review?

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2) Why it is important for the researcher to keep a record of the material collected?

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5.5 WRITING REVIEW OF LITERATURE

Writing literature review is an important part of the research process. According to Priest (2010), “putting the existing literature together in summary form, showing how it relates to a new problem (or a new interpretation of an old one) and identifying gaps and opportunities in the existing research record is both demanding and creative”.

There is no single way of writing review of literature as it can be written in different ways. Researchers generally begin with a discussion of their theoretical framework and literature review before they develop their research questions. However, according to Brennen (2013), “some qualitative researchers interact with previously published literature throughout their reports, forming an extended conversation with other researchers”.

5.5.1 Organisation of the Material

Material collected can be organised in the following ways:

- Chronological
- Thematic
- Methodological
- Area-wise

Chronological: Writing a review of literature in chronological manner is one of the most common ways of documentation in which you start from the earliest published studies then gradually move on to the more recent ones. The emphasis here is on progression of work in a given area over a period of time. To take an example, if you are researching the topic Communication Strategies for HIV/AIDS, you will identify studies ever since the epidemic emerged in mid 80s in different parts of the world and then move on to the recent ones. Such

chronological classification will be linear in nature and may not take into account varied emergent themes investigated by different scholars and thus may lack continuity in terms of themes.

Thematic: During material collection some themes, issues and concepts are likely to emerge frequently. These emergent themes will have to be refined and merged together for the purpose of analysis. To extend the previous example of Communication Strategies for HIV/AIDS; some emergent themes could be: health communication; HIV as a gender issue; role of mass media; role of alternate media, among others. Here the organisation of material is around a theme rather than progression over a period of time. However, even then, chronological development of these themes may be required to provide a flow and structuring of the content.

Methodological: The studies may also be organised in terms of theoretical frameworks, methods and materials used and key variables investigated. For example, you may organise the studies in terms of qualitative and quantitative research and mixed method approaches. This will enable you to assess their methodological strengths and weaknesses and reveal how research practices differ across groups, times and settings.

Area-wise: Material can also be classified in relation to the part of the world from where the studies emanate, for example, Communication Strategies for HIV/AIDS in Developing Countries; Developed Countries; Eastern Europe & CIS; and Studies in India. Such a categorisation can trace common emergent trends from a specific geo-political area for the purpose of analysis. Many developing countries are likely to share similar set of problems and concerns and discussing them in broad terms can reveal common patterns and findings.

From the above analysis you would have seen that the review of literature follows many patterns and approaches. You can also select one way say, theme-wise and then proceed to include studies in a chronological manner - starting from the early studies to the latest ones. Each theme should be given a sub-head and within that you may proceed chronologically. The bottom line is that there should be some rationale and justification for placing the studies under a broad head. A thread should run throughout the entire review so that the diverse studies are linked in some way to reveal emergent patterns and trends.

5.5.2 Writing the Reviews

Now that the material collected for literature review has been organised following a pattern, it is time for writing. Literature review, like any academic work is written in a formal style avoiding colloquial words and expressions. It has an introduction, a body and a summary. The introduction sets the context and guides the reader what to expect - the broad themes and areas identified for discussion. The body discusses the focus of each study, the methods and materials used, the major findings and recommendations while the summary provides a gist of your conclusions and the argument you wish to make. Let us discuss this in some detail.

Introduction: The Review of Literature in a thesis or dissertation generally comes after introduction and theoretical framework of the study (if discussed separately). At times, it is also made a part of the introductory chapters. Whatever approach

you take, the first few paragraphs should provide a gist to the reader what to expect in the review in a crisp and concise manner. A brief introduction introduces the central theme and establishes connections between various themes of the studies discussed in sections and sub-sections.

The Body: This part of the report summarises the findings of all the relevant research work that has already been undertaken. One of the common practices in writing review is to present a synthesis of the work undertaken by an author. This is generally done by selecting the most important point in each source, outlining the problem examined, describing the research methods and tools used. Towards the end, major research findings are summarised and suggestions for future research enunciated. It is possible that some studies may not provide all these details and as researchers, you will have to decide to include them in the review or not.

According to Berger (2000), 'Review of Literature in essence is a form of editing, in which quotations and sometimes summaries, phrases and synthesis of the material read from various scholars, authors and researchers are collected to produce an essay or article that makes its argument'. While writing you may use original sentences, phrases and expressions and some short quotations of the respective authors/ researchers for infusing individuality.

The review can be written either in the present tense or past tense but consistency should be maintained. Instead of 'the author says', verbs such as argues, examines, explores, claims, emphasises, observes, stresses, states etc. may be used. Some linking words such as, in addition, similarly, moreover, however, etc. may be used. To highlight a different point of view you may write - striking a different note, on the other hand, in a contrary view, nevertheless, etc.

Summary: After capturing the essence of the entire gamut of studies in a systematic manner, you have to summarise their major thrust areas and recommendations in two-three paragraphs. Thus you are summarising and synthesising within each source as well as throughout the review. Here, you also need to enumerate the strengths and limitations of the studies and the gaps revealed in them. The strengths of the studies could be in terms of choosing varied areas for exploration, rigour in the research design, use of multiple tools of data collection and refined statistical tools for data analysis, path-breaking findings, among others. The limitations of the study could be weak research design, inadequate sample size, lack of focus etc.

Ideally, your study should aim to fill some kind of gap traced during the review and the summary should 'make an argument'. Like any good piece of writing you should edit the content and revise and re-revise to firm up the section/ chapter on review.

5.5.3 Ethical issues

While writing a review, you also need to keep various ethical issues in mind. For example you should present the ideas of other scholars accurately and be respectful of others' opinions. You should give due credit and proper citations from wherever you source the material. Citation is an important 'intellectual tool' for the researchers. According to Bertrand & Hughes (2013),

- “It adds credibility to your research by providing evidence that you have consulted other sources.
- It enables you to share your research with other researchers. The citations of others have provided valuable sources of information to you, and your citations are part of a reciprocal system of information exchange.
- Correct and accurate citation ensures that you cannot be accused of plagiarism”.

You should not give any slant to the findings of other researchers and edit the material to show your own work in a positive light. You should also adhere to the copyright law and not use more than 60 words at a stretch and put material in quotations and cite the source.

While reviewing, you may come across some works which may be very relevant for your study. However, you should not attempt to copy verbatim and pass off the data as your own as you will get into plagiarism charges. Plagiarism is the unacknowledged passing of someone else’s words or ideas as your own without proper acknowledgment of your sources (Bertrand & Hughes, 2013). Now plagiarism has become an extremely serious matter and various anti plagiarism softwares are available in this regard. At the Indira Gandhi National Open University, it is mandatory to run each Ph.D. thesis submitted by a scholar through anti plagiarism software to address this menace.

Activity-2

Go through the studies identified in Activity-I closely. Analyse their major findings and trace the emergent trends.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) Among different ways of classification of material which one will you select and why?

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- 2) List three points which you will keep in view while writing review of literature.

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5.6 LET US SUM UP

In this unit, we looked at various aspects of Review of Literature and tried to demystify the topic. We started with discussing that review of literature is a form of secondary research which reviews the available research studies in a given area/ topic. The major works are identified, classified and summarised to make an argument. It is not review of great literary texts on which personal opinion on the materials read has to be provided or summary of available bibliographical material and citation of various sources at one place.

We discussed that review helps in refining your ideas and concepts, firms up your objectives and hypotheses. It also helps in selecting appropriate theoretical framework, research methods, tools and techniques for your study. A critical review of existing materials helps to determine which type of research, objectives, themes and topics have been over emphasised, under emphasised and neglected and what more can be done to make research more comprehensive.

We described the process of material selection, collection and organisation and suggested that research studies for reviews may be classified chronologically, thematically and methodologically or in a combination of two.

The process of writing review of literature was discussed in detail. It is generally done by selecting the most important point in each source, outlining the problem examined, describing the research methods and tools used and summarising the major research findings and giving suggestions for future research. It was emphasised that through such analysis the literature review identifies some gap which your study aims to fill.

Towards the end, some ethical issues were briefly discussed that ideas of other scholars should be presented accurately with due credit and proper citations. It was stressed that to show your own work in a positive light, you should not give any slant to the findings of other researchers and edit the material indiscriminately.

It is hoped that the above analysis will help you in reviewing literature for your research. This will help you approach, classify and write the review using your own style and contribute to the scholarship in your discipline.

5.7 REFERENCES AND FURTHER READINGS

Berger, Arther Asa (2000), Media and Communication Research Methods, Sage Publications, London, New Delhi

Brennen Bonnie S. (2013), Qualitative Research Methods for Media Studies, Routledge, New York

Bertrand Ina & Peter Hughes (2005), Media Research Methods: Audiences, Institutions, Texts, Palgrave MacMillan, New York

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Wimmer R.D. & Joseph R. Dominick (2004), Mass media Research: An Introduction, 4th edition, California, Wadsworth Series in Mass Communication

5.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) The theoretical framework guides and links a study with the larger body of work available in a specific area. It systematically and logically attempts to explain the phenomenon being studied.
- 2) The research objectives need to be stated clearly in behavioural terms so that they are observable and measurable. Some examples of behavioural terms are: define, describe, explain, state, compare, differentiate, identify, distinguish, summarise, illustrate, outline etc.

Check Your Progress 2

- 1) While selecting material for literature review, I shall keep the following points in mind:
 - a) The amount of information should be adequate for the study.
 - b) The level of information should be appropriate for academic work.
 - c) The information selected should be specific, in-depth and not cursory or superficial in nature.
- 2) It is useful to keep a record of the material collected as it may be needed at the time of writing research report for cross-checking as well as citations. The date on which the material is retrieved from website, the names of authors especially the foreign authors, places of publications etc. should be clearly recorded.

Check Your Progress 3

- 1) I will select a theme-wise classification of material which will include studies written in a chronological manner. I shall give each theme a sub-head and proceed chronologically - starting from early studies moving on to the more recent ones. This way, I shall provide some rationale and justification for placing the studies under a specific head.
- 2) While writing review of literature, I shall briefly introduce the subject and how the material has been classified and presented. Then I shall take up the studies and outline their major thrust areas, methodological approach followed, major findings and recommendations. Towards the end, I shall provide the summary of the various studies - outlining their strengths and limitations and how my work will try to bridge the gap traced in these studies.

